ALGEBRA

बीजगणित

SHEET-01

CLASS NOTES

- Algebra
- Trigo
- No. System
- Geo
- Mens → 2D
  - Mens → 3D

- Basic
- Formula (a^3+b^3)
- Symmetricity
- Value putting
- Misc.

Type-01

$\left(x+\frac{1}{x}\right)$ $\left(x-\frac{1}{x}\right)$

$x+\frac{1}{x}=n$

$$\left(x+\frac{1}{x}\right)^2=(n)^2$$

$$\Rightarrow x^2+\frac{1}{x^2}+2\cancel{x}\cancel{x}\times\frac{1}{\cancel{x}}=n^2$$

$$\therefore x^2+\frac{1}{x^2}=n^2-2$$

$$(a+b)^2=a^2+b^2+2ab$$

$$x + \frac{1}{x} = n$$

- $x^2 + \frac{1}{x^2} = n^2 - 2$

- $x^4 + \frac{1}{x^4} = (n^2 - 2)^2 - 2$

Q. $x+\frac{1}{x}=3$

Ans $x^2+\frac{1}{x^2}=3^2-2=7$

$x^4+\frac{1}{x^4}=7^2-2=47$

$x^8+\frac{1}{x^8}=47^2-2$

$=2209-2$

$=2207$

Q. $x+\frac{1}{x}=4$

Ans $x^2+\frac{1}{x^2}=4^2-2=14$

$x^4+\frac{1}{x^4}=14^2-2=194$

$x^8+\frac{1}{x^8}=(194)^2-2$

$=$ _____

$x+\frac{1}{x}=n$

- $x^2+\frac{1}{x^2}=n^2-2$
- $x^3+\frac{1}{x^3}=n^3-3n$

⊙ $x+\frac{1}{x}=3$

- $x^3+\frac{1}{x^3}=n^3-3n$
  $=27-9=\boxed{18}$
- $x^6+\frac{1}{x^6}=18^2-2=\underline{322}$

$$x+\frac{1}{x}=n$$

- $x^2+\frac{1}{x^2} = n^2-2$
- $x^3+\frac{1}{x^3} = n^3-3n$
- $x^4+\frac{1}{x^4} = (n^2-2)^2-2$
- $x^5+\frac{1}{x^5} = (n^2-2)(n^3-3n)-n$
- $x^6+\frac{1}{x^6} = (n^3-3n)^2-2$
- $x^7+\frac{1}{x^7} = \left(x^3+\frac{1}{x^3}\right)\left(x^4+\frac{1}{x^4}\right)-\left(x+\frac{1}{x}\right)$

$\left(x^2+\frac{1}{x^2}\right)\left(x^3+\frac{1}{x^3}\right)-\left(x+\frac{1}{x}\right)$

If $x + \frac{1}{x} = n$

- $x^2 + \frac{1}{x^2} = n^2 - 2$
- $x^3 + \frac{1}{x^3} = n^3 - 3n$
- $x^4 + \frac{1}{x^4} = (n^2 - 2)^2 - 2$
- $x^5 + \frac{1}{x^5} = (n^2 - 2)(n^3 - 3n) - n$
- $x^6 + \frac{1}{x^6} = (n^3 - 3n)^2 - 2$

If $x - \frac{1}{x} = n$

- $x^2 + \frac{1}{x^2} = n^2 + 2$
- $x^3 - \frac{1}{x^3} = n^3 + 3n$
- $x^4 + \frac{1}{x^4} = (n^2 + 2)^2 - 2$
- $x^5 - \frac{1}{x^5} = (n^2 + 2)(n^3 + 3n) - n$
- $x^6 + \frac{1}{x^6} = (n^2 + 3n)^2 + 2$

1. If $x + \frac{1}{x} = 3$ Find

(i) $x^2 + \frac{1}{x^2}$

(ii) $x^3 + \frac{1}{x^3}$

(iii) $x^4 + \frac{1}{x^4}$

(iv) $x^5 + \frac{1}{x^5}$

(v) $x^6 + \frac{1}{x^6}$

(vi) $x^7 + \frac{1}{x^7}$

1. If $x + \frac{1}{x} = 3$ Find

(i) $x^2 + \frac{1}{x^2}$

$= n^2 - 2$

$= 3^2 - 2$

$= 7$

1. If $x + \frac{1}{x} = 3$ Find

(ii) $x^3 + \frac{1}{x^3}$

$= n^3 - 3n$

$= 27 - 9$

$= 18$

$$x + \frac{1}{x} = 3$$

- $x^2 + \frac{1}{x^2} = 3^2 - 2 = 7$
- $x^3 + \frac{1}{x^3} = 3^3 - 3 \times 3 = 18$
- $x^4 + \frac{1}{x^4} = 7^2 - 2 = 44$
- $x^5 + \frac{1}{x^5} = 7 \times 18 - 3 = 126 - 3 = 123$
- $x^6 + \frac{1}{x^6} = (18)^2 - 2 = 322$
- $x^7 + \frac{1}{x^7} = 18 \times 44 - 3 = 846 - 3 = 843$

2. If $x - \frac{1}{x} = 4$, Find

$x^6 + \frac{1}{x^6} = (76)^2 + 2$
$= 5776 + 2$
$= 5778$

(i) $x^2 + \frac{1}{x^2} = n^2 + 2$
$= 18$

Hold

(ii) $x^2 - \frac{1}{x^2}$

(iii) $x^3 - \frac{1}{x^3} = n^3 + 3n$
$= 64 + 12$
$= 76$

(iv) $x^4 + \frac{1}{x^4}$
$= (18)^2 - 2$
$= 322$

(v) $x^5 - \frac{1}{x^5} = 18 \times 76 - 4$
$= 1368 - 4$
$= 1364$

(vi) $x^6 + \frac{1}{x^6}$

∴

① $x-\frac{1}{x}=4$ $\qquad$ $x^2-\frac{1}{x^2}=$

Ans $x-\frac{1}{x}=4$

$$x+\frac{1}{x}=\sqrt{4^2+4}=\sqrt{20}=2\sqrt{5}$$

$$\therefore x^2-\frac{1}{x^2}=\left(x-\frac{1}{x}\right)\left(x+\frac{1}{x}\right)$$

$$=4\times 2\sqrt{5}$$

$$=8\sqrt{5}$$ ✓

3. If $x + \frac{1}{x} = 7$, find the value of $x^3 + \frac{1}{x^3}$.

SSC Phase-XII 25/06/2024 (Shift-04)

(a) 343 (b) 332

(c) 340 (d) 322

$x^3 + \frac{1}{x^3} = n^3 - 3n$

$= 343 - 21$

$= 322$

4. If $a + \frac{1}{a} = 3$, then the value of $a^4 + \frac{1}{a^4}$ is:

SSC CPO 04/10/2023 (Shift-01)

(a) 27 (b) 81

(c) 48 (d) 47

$a + \frac{1}{a} = 3$

$a^4 + \frac{1}{a^4} = (7^2-2)^2 - 2$

$= (9-2)^2 - 2$

$= 49 - 2$

$= 47$

$a + \frac{1}{a} = 3$

$a^2 + \frac{1}{a^2} = 3^2 - 2 = 7$

$a^4 + \frac{1}{a^4} = 7^2 - 2 = 47$

4. If $a + \frac{1}{a} = 3$, then the value of $a^4 + \frac{1}{a^4}$ is:

SSC CPO 04/10/2023 (Shift-01)

(a) 27+2 = 29

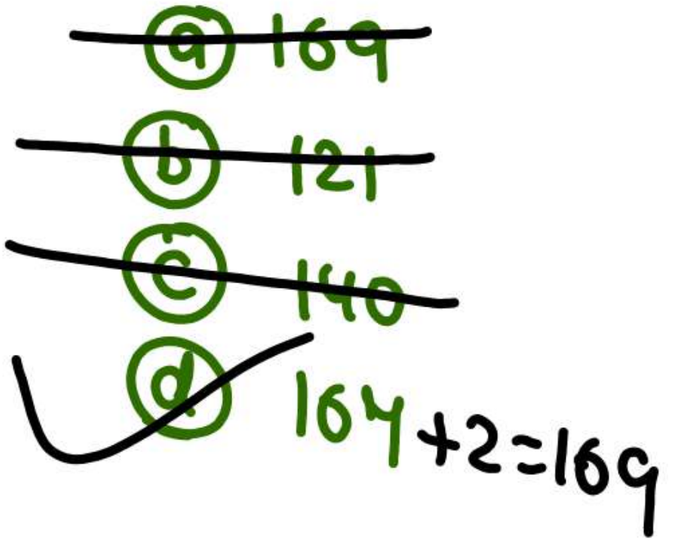
(b) 81+2 = 83

(c) 48+2 = 50

(d) 47+2 = 49

Q. $x + \frac{1}{x} = 13$

$x^2 + \frac{1}{x^2} = ?$



Q. $x + \frac{1}{x} = 52$

$x^2 + \frac{1}{x^2} = ?$

(a) 2404
(b) 2500
(c) 2400
(d) 2402

2406
2502
2402
2404
(52)2

5. If $a+\frac{1}{a}=7$, then $a^5+\frac{1}{a^5}$ is equal to:

SSC CGL 17/07/2023 (Shift-02)

(a) 15127 (b) 13127

(c) 14527 (d) 11512

$$a^5+\frac{1}{a^5}=(n^2-2)(n^3-3n)-n$$

$$=44\times322-4$$

$$=15134-4$$

$$=15124$$

5. If $a+\frac{1}{a}=7$, then $a^5+\frac{1}{a^5}$ is equal to:

SSC CGL 17/07/2023 (Shift-02)

(a) 15127 (b) 13127

(c) 14527 (d) 11512

$a^5+\frac{1}{a^5}=(n^2-2)(n^3-3n)-n$

$=47\times322-7$

D.S

$2\times7-7$

$14-7$

7

6. If $x+\frac{1}{x}=-6$, what will be the value of

$$x^5+\frac{1}{x^5}$$

SSC CGL 17/07/2023 (Shift-03)

(a) –7776 (b) –6726

(c) –6730 (d) –6732

$x^5+\frac{1}{x^5} = (n^2-2)(n^3-3n)-n$

$= (36-2)(-216+18)-(-6)$

$= 34\times(-198)+6$

$= -6732+6$

$= -6726$

$a+\frac{1}{a}=7\sqrt{3}$

$a^6+\frac{1}{a^6}=(n^3-3n)^2-2$

$=\left(1029\sqrt{3}-21\sqrt{3}\right)^2-2$

$=\left(1008\sqrt{3}\right)^2-2$

$=1016064\times3-2$

$=3048192-2$

$=3048190$

7. If $\left(a+\frac{1}{a}\right)=7\sqrt{3}$, then what is the value of (a^6+a^{-6})?

SSC CGL 19/09/2024 (Shift-01)

(a) 3048190 ✓
(b) 3048542
(c) 3048132
(d) 3048625

* $a+\frac{1}{a}=n$

$a^3+\frac{1}{a^3}=n^3-3n$

$a^6+\frac{1}{a^6}=(n^3-3n)^2-2$

7. If $\left(a+\frac{1}{a}\right)=7\sqrt{3}$, then what is the value of $(a^6 + a^{-6})$?

SSC CGL 19/09/2024 (Shift-01)

(a) 3048190
(b) 3048542
(c) 3048132
(d) 3048625

$a^6+\frac{1}{a^6} = (n^3-3n)^2-2$

$= (1029\sqrt{3}-21\sqrt{3})^2-2$

$= (1008\sqrt{3})^2-2$

$= 1016064\times3-2$

$=$

$2-2=0$

$a^{-1}=\frac{1}{a}$

$a^{-2}=\frac{1}{a^2}$

$a^{-6}=\frac{1}{a^6}$

$x + \frac{1}{x} = n$

* $x^2 + \frac{1}{x^2} = n^2 - 2$

* $x^3 + \frac{1}{x^3} = n^3 - 3n$

* $x^4 + \frac{1}{x^4} = (n^2-2)^2 - 2$

* $x^5 + \frac{1}{x^5} = (n^2-2)(n^3-3n) - n$

* $x^6 + \frac{1}{x^6} = (n^3-3n)^2 - 2$

$x - \frac{1}{x} = n$

* $x^2 + \frac{1}{x^2} = n^2 + 2$

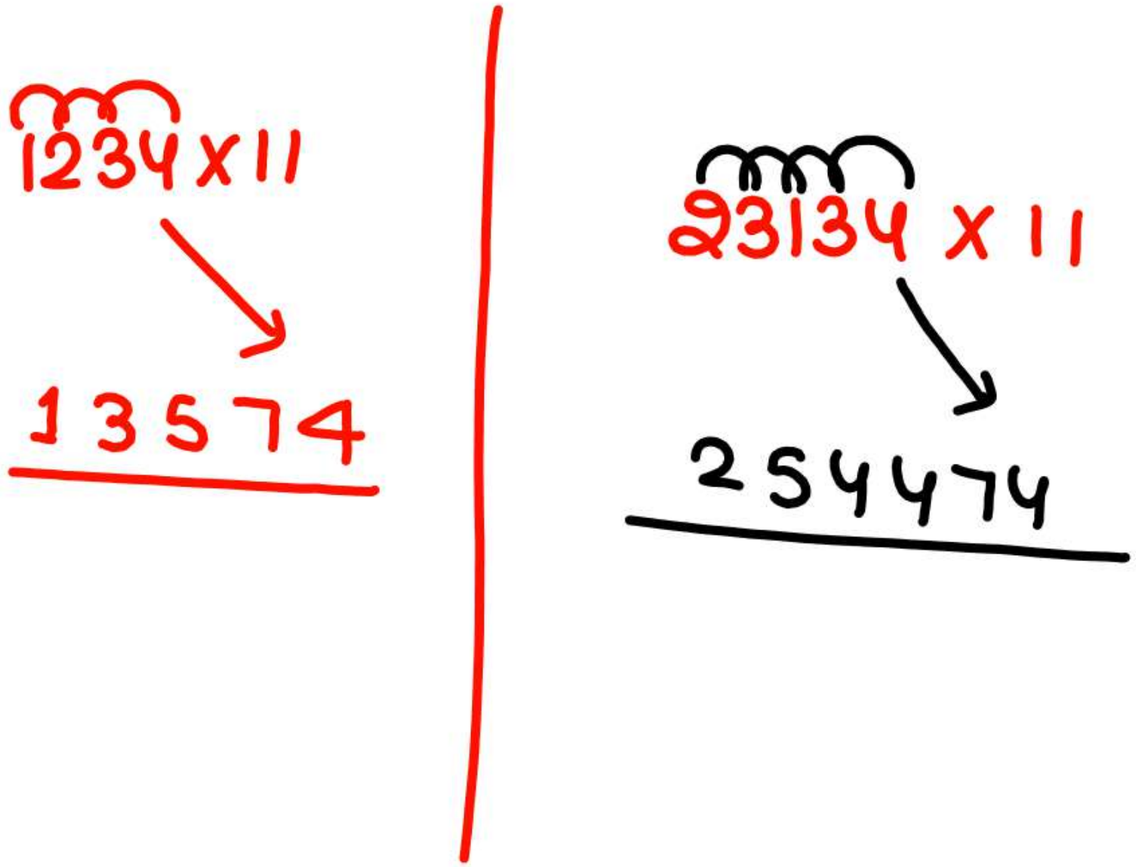
* $x^3 - \frac{1}{x^3} = n^3 + 3n$

* $x^4 + \frac{1}{x^4} = (n^2+2)^2 - 2$

* $x^5 - \frac{1}{x^5} = (n^2+2)(n^3+3n) - n$

* $x^6 + \frac{1}{x^6} = (n^3+3n)^2 + 2$ *

⋮


1234 x 11
13574
23134 x 11
254474

$$x - \frac{1}{x} = 3$$

* $x^2 + \frac{1}{x^2} = n^2 + 2 = 11$

* $x^3 - \frac{1}{x^3} = n^3 + 3n = 24 + 9 = 36$

* $x^4 + \frac{1}{x^4} = (11)^2 - 2 = 119$

* $x^5 - \frac{1}{x^5} = 11 \times 36 - 3$
$= 396 - 3 = 393$

* $x^6 + \frac{1}{x^6} = (36)^2 + 2$
$= 1296 + 2 = 1298$

$x - \frac{1}{x} = 10$

$x^2 + \frac{1}{x^2} = 10^2 + 2 = 102$

$x^4 + \frac{1}{x^4} = (102)^2 - 2$

$= 10404 - 2$

$= 10402$

8. If $\left(x - \frac{1}{x}\right) = 10$, what is the value of $x^4 + \frac{1}{x^4}$?

SSC CGL 26/07/2023 (Shift-03)

(a) 10404
(b) 10402
(c) 10406
(d) 10400

$(102)^2$	$(103)^2$	$(104)^2$
10404	10609	10816

9. If $x - \frac{1}{x} = 10$, then $x^3 - \frac{1}{x^3}$ is equal to :

SSC CHSL 04/07/2019 (Shift-02)

(a) 970
(b) 1000
(c) 1030
(d) 1100

$x - \frac{1}{x} = n$ (10)

$x^3 - \frac{1}{x^3} = n^3 + 3n$

$= 1000 + 30$

$= 1030$

$3\left(y-\frac{1}{y}\right)=5$

$y-\frac{1}{y}=\frac{5}{3}$

$y^2+\frac{1}{y^2}=n^2+2$

$=\frac{25}{9}+2$

$=\frac{43}{9}$

10. If $\left(3y-\frac{3}{y}\right)=5$, find the value of $\left(y^2+\frac{1}{y^2}\right)$.

* SSC CGL 19/07/2023 (Shift-02)

(a) $\frac{47}{9}$

(b) $\frac{49}{9}$

(c) $\frac{41}{9}$

(d) $\frac{43}{9}$

$x - \frac{1}{x} = 13$

$x^2 + \frac{1}{x^2} = (13)^2 + 2 = 171$

$x^4 + \frac{1}{x^4} = (171)^2 - 2$

$= \ldots 1 - 2$

$= \ldots 9$

11. If $x - \frac{1}{x} = 13$, what will be the value of $x^4 + \frac{1}{x^4}$?

SSC CGL 03/12/2022 (Shift-01)

(a) 28561
(b) 29243
(c) 27887
(d) 29239

$29 - 8 = 13$

$x-\frac{1}{x}=n$

$x^5-\frac{1}{x^5}=(n^2+2)(n^3+3n)-n$

$=27\times140-5$

$=3780-5$

$=3775$

12. If $\left(x-\frac{1}{x}\right)=5$, then the value of $\left(x^5-\frac{1}{x^5}\right)$ is :

SSC CHSL 07/08/2023 (Shift-02)

(a) 3775 ✓

(b) 3740

(c) 3715

(d) 3725

13. If $x - \frac{1}{x} = -6$, what will be the value of $x^5 - \frac{1}{x^5}$?

SSC CGL 18/07/2023 (Shift-02)

(a) – 8898 (b) – 8896

(c) – 8886 (d) – 8892

Ans = $(36+2)(-216-18)-(-6)$

$= 38 \times (-234) + 6$

$= -8892 + 6$

$= -8886$

$x - \frac{1}{x} = n$

$x^5 - \frac{1}{x^5} = (n^2+2)(n^3+3n) - n$

-6

Ans = $(64+12)^2+2$

$= (76)^2+2$

$= \ldots 6+2$

14. If $\left(y-\frac{1}{y}\right)=4$, find the value of $\left(y^6+\frac{1}{y^6}\right)$.

SSC CGL 25/07/2023 (Shift-01)

(a) 5774 (b) 4096

(c) 5776 (d) 5778

$y-\frac{1}{y}=4$

$y^6+\frac{1}{y^6}=(n^3+3n)^2+2$

4

15. If $x > 1$ and $x^2 + \frac{1}{x^2} = 83$ then, $x^3 - \frac{1}{x^3}$ is:

Hold

SSC CGL 17/09/2024 (Shift-01)

(a) 884
(b) 876
(c) 754
(d) 756

Q. $x^{1} + \frac{1}{x} = 4$

$x^{2} + \frac{1}{x^{2}} = 4^{2} - 2$

$= 44$

Q. $x^{2} + \frac{1}{x^{2}} = 4$

$x^{4} + \frac{1}{x^{4}} = 7^{2} - 2$

$= 47$

Q. $x^{100} + \frac{1}{x^{100}} = 9$

$x^{200} + \frac{1}{x^{200}} = 9^2 - 2$

$= 79$

Q. $\left(\frac{a}{b}\right)^1 + \left(\frac{b}{a}\right)^1 = 5$

$\left(\frac{a}{b}\right)^2 + \left(\frac{b}{a}\right)^2 = 25 - 2$

$= 23$

Q $\sqrt{x} + \frac{1}{\sqrt{x}} = 6$

$x + \frac{1}{x} = ?$

Ans $(x)^{\frac{1}{2}} + \frac{1}{(x)^{\frac{1}{2}}} = 6$

$x^1 + \frac{1}{x} = \underline{34}$

Q. $\sqrt{x} + \frac{1}{\sqrt{x}} = 3$

Ans $x + \frac{1}{x} = 3^2 - 2 = 7$

$x^2 + \frac{1}{x^2} = 7^2 - 2 = 47$

$x^4 + \frac{1}{x^4} = (47)^2 - 2$

$= 2209 - 2$

$= \underline{2207}$

Q. $x\sqrt{x} + \frac{1}{x\sqrt{x}} = 5$

Ans $x^3 + \frac{1}{x^3} = n^2 - 2 = 5^2 - 2 = 23$

$x\sqrt{x} = x^1 \times x^{\frac{1}{2}} = x^{1+\frac{1}{2}} = x^{\frac{3}{2}}$) $\times 2$

x^3)

$$\left(\sqrt{x}+\frac{1}{\sqrt{x}}\right)^2=(7)^2$$

$$x+\frac{1}{x}+2\times\sqrt{x}\times\frac{1}{\sqrt{x}}=49$$

$$x+\frac{1}{x}=49-2=44$$

16. If $\sqrt{x}+\frac{1}{\sqrt{x}}=7$, then the value of $x+\frac{1}{x}$ is equal to:

SSC CGL 10/09/2024 (Shift-02)

(a) 47
(b) 49
(c) 51
(d) 45

16. If $\sqrt{x} + \frac{1}{\sqrt{x}} = 7$, then the value of $x + \frac{1}{x}$ is equal to:

SSC CGL 10/09/2024 (Shift-02)

(a) $47+2=49$
(b) $49+2=51$
(c) $51+2=53$
(d) $45+2=47$

Basic

$\left(\sqrt{x}-\frac{1}{\sqrt{x}}\right)^2=(4)^2$

17. If $\sqrt{x}-\frac{1}{\sqrt{x}}=4$, then $x^2+\frac{1}{x^2}$ is equal to :

SSC CGL 10/06/2019 (Shift-01)

(a) 192
(b) 326
(c) 322
(d) 256

$x+\frac{1}{x}-2\times\sqrt{x}\times\frac{1}{\sqrt{x}}=16$

$x+\frac{1}{x}=16+2=18$

$x^2+\frac{1}{x^2}=18^2-2=322$

$(a-b)^2=a^2+b^2-2ab$

$\sqrt{x} - \frac{1}{\sqrt{x}} = 4$

17. If $\sqrt{x} - \frac{1}{\sqrt{x}} = 4$, then $x^2 + \frac{1}{x^2}$ is equal to :

SSC CGL 10/06/2019 (Shift-01)

(a) 192
(b) 326
(c) 322
(d) 256

$x + \frac{1}{x} = 4^2 + 2 = 18$

$x^2 + \frac{1}{x^2} = 18^2 - 2 = 322$

$(a-b)^2 = a^2 + b^2 - 2ab$

17. If $\sqrt{x} - \frac{1}{\sqrt{x}} = 4$, then $x^2 + \frac{1}{x^2}$ is equal to :

SSC CGL 10/06/2019 (Shift-01)

(a) $192+2=194$ ✗ ~~(b) 326~~

(c) $322+2=324$ ✓ ~~(d) 256~~

18. If $\sqrt{x} - \frac{1}{\sqrt{x}} = \sqrt{6}$, then $x^2 + \frac{1}{x^2}$ is equal to :

SSC CGL 11/06/2019 (Shift-02)

(a) 62+2=64

(b) 40

(c) 54

(d) 66

$\left(\sqrt{x}-\frac{1}{\sqrt{x}}\right)^2=(\sqrt{6})^2$

$x+\frac{1}{x}-2\times\sqrt{x}\times\frac{1}{\sqrt{x}}=6$

$x+\frac{1}{x}=8$

$x^2+\frac{1}{x^2}=8^2-2=62$

18. If $\sqrt{x}-\frac{1}{\sqrt{x}}=\sqrt{6}$, then $x^2+\frac{1}{x^2}$ is equal to :

SSC CGL 11/06/2019 (Shift-02)

(a) 62 (b) 40

(c) 54 (d) 66

Bridging

* If $x + \frac{1}{x} = n$ $\quad x - \frac{1}{x} = \sqrt{n^2 - 4}$

* If $x - \frac{1}{x} = n$ $\quad x + \frac{1}{x} = \sqrt{n^2 + 4}$

Q. If $x+\frac{1}{x}=4$ $\quad x-\frac{1}{x}=\sqrt{n^2-4}=\sqrt{12}=2\sqrt{3}$

Q. If $x+\frac{1}{x}=5$ $\quad x-\frac{1}{x}=\sqrt{n^2-4}=\sqrt{21}$ ✓

Q. If $x-\frac{1}{x}=3$ $\quad x+\frac{1}{x}=\sqrt{n^2+4}=\sqrt{13}$

Q. If $x-\frac{1}{x}=6$ $\quad x+\frac{1}{x}=\sqrt{n^2+4}=\sqrt{40}=\underline{2\sqrt{10}}$

$(\sqrt{7})^3 = 7\sqrt{7}$

$(\sqrt{5})^3 = 5\sqrt{5}$

Q. $x + \frac{1}{x} = 5$

$x^3 - \frac{1}{x^3} =$

Ans $x - \frac{1}{x} = \sqrt{n^2 - 4} = \sqrt{21}$ (n)

$x^3 - \frac{1}{x^3} = n^3 + 3n$

$= 21\sqrt{21} + 3\sqrt{21}$

$= 24\sqrt{21}$ ✓

✗ (a) 110

✗ (b) 125

✗ (c) 140

(d) N.O.T

CONCEPT OF BRIDGING

If $x - \frac{1}{x} = n$ then, $x + \frac{1}{x} = \sqrt{n^2 + 4}$

If $x + \frac{1}{x} = n$ then, $x - \frac{1}{x} = \sqrt{n^2 - 4}$

19. If $\left(x+\frac{1}{x}\right)=7$, $x>0$. The positive value of $\left(x-\frac{1}{x}\right)$ is:

$x-\frac{1}{x}=\sqrt{m^2-4}$

$=\sqrt{49-4}=\sqrt{45}$

$=\sqrt{5\times 9}$

$=3\sqrt{5}$

SSC CGL 19/09/2024 (Shift-02)

(a) $5\sqrt{4}$

(b) $5\sqrt{5}$

(c) $3\sqrt{5}$

(d) $4\sqrt{5}$

$(88\underline{5})^2 = 7832\underline{25}$

$(15)^2 =$

$(25)^2 = 625$

$(35)^2 = 1225$

$(85)^2 = 7225$

$(95)^2 = \underline{9025}$

$(105)^2$

$(11\underline{5})^2 = 132\underline{25}$

20. If x is an integar such that $x + \frac{1}{x} = \frac{65}{8}$, then find the value of $x - \frac{1}{x}$.

$= \sqrt{m^2-4}$

* SSC CGL 17/09/2024 (Shift-03)

(a) $\frac{63}{6}$

(b) $\frac{63}{8}$

(c) $\frac{36}{8}$

(d) $\frac{32}{6}$

$a^2-b^2=(a-b)(a+b)$

Ans $= \sqrt{\left(\frac{65}{8}\right)^2-(2)^2}$

$= \sqrt{\left(\frac{65}{8}-2\right)\left(\frac{65}{8}+2\right)}$

$= \sqrt{\frac{49}{8}\times\frac{81}{8}}$

$= \frac{7\times9}{8} = \frac{63}{8}$

① $x + \frac{1}{x} = 4$

$\sqrt{n^2-4}$ $\quad$ n

$x^2 - \frac{1}{x^2} = \left(x - \frac{1}{x}\right)\left(x + \frac{1}{x}\right)$

$= \sqrt{4^2-4} \times 4$

$= \sqrt{12} \times 4$

$= 2\sqrt{3} \times 4$

$= \underline{8\sqrt{3}}$

$(a^2-b^2) = (a-b)(a+b)$

Q. $x+\frac{1}{x}=5$ $\qquad x^2-\frac{1}{x^2}=?$

Ans $x^2-\frac{1}{x^2}=\left(x-\frac{1}{x}\right)\left(x+\frac{1}{x}\right)$

$=\sqrt{5^2-4}\qquad 5$

$=\sqrt{21}\times 5$

$=5\sqrt{21}$

$a^2-b^2=(a-b)(a+b)$

Q. $x+\frac{1}{x}=10$

$x-\frac{1}{x}=?$

Ans $x-\frac{1}{x}=\sqrt{m^2-4}$

$=\sqrt{100-4}$

$=\sqrt{96}$

$=\sqrt{16\times6}$

$=\underline{4\sqrt{6}}$

Q. $x+\frac{1}{x}=8$

$x-\frac{1}{x}=?$

Ans $x-\frac{1}{x}=\sqrt{m^2-4}$

$=\sqrt{64-4}$

$=\sqrt{60}$

$=\sqrt{4\times15}$

$=\underline{2\sqrt{15}}$

Q. $x+\frac{1}{x}=10$

$x^2-\frac{1}{x^2}=?$

Ans $x-\frac{1}{x}=\sqrt{10^2-4}=\sqrt{96}$

$=4\sqrt{6}$

$x^2-\frac{1}{x^2}=\left(x-\frac{1}{x}\right)\left(x+\frac{1}{x}\right)$

$=4\sqrt{6}\times 10$

$=40\sqrt{6}$

Q. $x+\frac{1}{x}=8$

$x^2-\frac{1}{x^2}=?$

Ans $\left(x^2-\frac{1}{x^2}\right)=\left(x-\frac{1}{x}\right)\left(x+\frac{1}{x}\right)$

$=\left(\sqrt{8^2-4}\right)\times 8$

$=2\sqrt{15}\times 8$

$=16\sqrt{15}$

$$* \ (a^8 - b^8) = (a^4+b^4)(a^2+b^2)(a+b)(a-b)$$

$$= (a^4+b^4)(a^2+b^2)(a^2-b^2)$$

$$= (a^4+b^4)(a^4-b^4)$$

$$= (a^8-b^8)$$

* $(a^2-b^2) = (a+b)(a-b)$

* $(a^4-b^4) = (a^2+b^2)(a+b)(a-b)$

* $(a^8-b^8) = (a^4+b^4)(a^2+b^2)(a+b)(a-b)$

* $(a^{16}-b^{16}) = (a^8+b^8)(a^4+b^4)(a^2+b^2)(a+b)(a-b)$

* $\left(x^2-\frac{1}{x^2}\right) = \left(x+\frac{1}{x}\right)\left(x-\frac{1}{x}\right)$

* $\left(x^4-\frac{1}{x^4}\right) = \left(x^2+\frac{1}{x^2}\right)\left(x+\frac{1}{x}\right)\left(x-\frac{1}{x}\right)$

* $\left(x^8-\frac{1}{x^8}\right) = \left(x^4+\frac{1}{x^4}\right)\left(x^2+\frac{1}{x^2}\right)\left(x+\frac{1}{x}\right)\left(x-\frac{1}{x}\right)$

* $\left(x^6-\frac{1}{x^6}\right) = \left(x^3+\frac{1}{x^3}\right)\left(x^3-\frac{1}{x^3}\right)$

$(a^{12}-b^{12}) = (a^6+b^6)(a^3+b^3)(a^3-b^3)$

Q. $x+\frac{1}{x}=3$ $\qquad$ $x^4-\frac{1}{x^4}=?$

$$\left(x^4-\frac{1}{x^4}\right)=\left(x^2+\frac{1}{x^2}\right)\left(x+\frac{1}{x}\right)\left(x-\frac{1}{x}\right)$$

$$=(3^2-2)\times 3\times\sqrt{3^2-4}$$

$$=7\times 3\times\sqrt{5}$$

$$=21\sqrt{5}$$

① $x+\frac{1}{x}=4$ $\qquad$ $x^8-\frac{1}{x^8}=?$

Ans $\left(x^8-\frac{1}{x^8}\right)=\left(x^4+\frac{1}{x^4}\right)\left(x^2+\frac{1}{x^2}\right)\left(x+\frac{1}{x}\right)\left(x-\frac{1}{x}\right)$

$= 194 \times 14 \times 4 \times \sqrt{16-4}$

$= 194 \times 56 \times 2\sqrt{3}$

$= 194 \times 112\sqrt{3}$

$= 21728\sqrt{3}$

Q. $x - \frac{1}{x} = 3$ $\qquad$ $x^8 - \frac{1}{x^8} = ?$

Ans $x^8 - \frac{1}{x^8} = \left(x^4 + \frac{1}{x^4}\right)\left(x^2 + \frac{1}{x^2}\right)\left(x + \frac{1}{x}\right)\left(x - \frac{1}{x}\right)$

$= 119 \times 11 \times \sqrt{13} \times 3$

$= 357 \times 11 \times \sqrt{13}$

$= 3927\sqrt{13}$

Q. $x+\frac{1}{x}=\sqrt{6}$ $\qquad$ $x^8-\frac{1}{x^8}$

Ans $x^8-\frac{1}{x^8}=\left(x^4+\frac{1}{x^4}\right)\left(x^2+\frac{1}{x^2}\right)\left(x+\frac{1}{x}\right)\left(x-\frac{1}{x}\right)$

$= \quad 14 \times 4 \times \sqrt{6} \times \sqrt{6-4}$

$= \quad 56\sqrt{6} \times \sqrt{2}$

$= \quad 56\sqrt{12}$

$= \quad 56 \times 2\sqrt{3}$

$= \quad 112\sqrt{3}$

Q. $x - \frac{1}{x} = \sqrt{6}$ $\qquad x^8 - \frac{1}{x^8} = ?$

Ans $\left(x^8 - \frac{1}{x^8}\right) = \left(x^4 + \frac{1}{x^4}\right)\left(x^2 + \frac{1}{x^2}\right)\left(x + \frac{1}{x}\right)\left(x - \frac{1}{x}\right)$

$= 62 \times 8 \times \sqrt{10} \times \sqrt{6}$

$= 496\sqrt{60}$

$= 496 \times 2 \times \sqrt{15}$

$= 992\sqrt{15}$

$x - \frac{1}{x} = \frac{4}{10} \frac{2}{5}$

. 21. If $\left(x - \frac{1}{x}\right) = 0.4$, and $x>0$, what is the value of $\left(x^2 - \frac{1}{x^2}\right)$?

SSC CHSL 07/08/2023 (Shift-02)

(a) $\frac{4}{25}\sqrt{26}$ (b) $\frac{2}{25}\sqrt{29}$

(c) $\frac{7}{25}\sqrt{26}$ (d) $\frac{3}{25}\sqrt{29}$

$\text{ans} = \left(x^2 - \frac{1}{x^2}\right) = \left(x - \frac{1}{x}\right)\left(x + \frac{1}{x}\right)$

$= \frac{2}{5} \times \sqrt{\left(\frac{2}{5}\right)^2 + 4}$

$= \frac{2}{5} \times \sqrt{\frac{4}{25} + 4}$

$= \frac{2}{5}\sqrt{\frac{104}{25}} = \frac{2}{5} \frac{\times 2\sqrt{26}}{\times 5} = \frac{4}{25}\sqrt{26}$

$\sqrt{104} = \sqrt{4 \times 26}$

22. If $x+\frac{1}{x}=2\sqrt{10}$ where $x>1$ then the value of $\left(x^3-\frac{1}{x^3}\right)$ is:

SSC CGL 11/09/2024 (Shift-01)

(a) 221 (b) 234
(c) 198 (d) 216

$\therefore x-\frac{1}{x}=\sqrt{(2\sqrt{10})^2-4}$

$=\sqrt{36}=6$

$\therefore x^3-\frac{1}{x^3}=6^3+3\times 6$

$=216+18=234$

$x-\frac{1}{x}=n$

$x^3-\frac{1}{x^3}=n^3+3n$

Q. $x + \frac{1}{x} = 4$

$x^2 - \frac{1}{x^2} = \left(x - \frac{1}{x}\right)\left(x + \frac{1}{x}\right)$

$= \sqrt{4^2 - 4} \quad 4$

$= \sqrt{12} \times 4$

$= 2\sqrt{3} \times 4$

$= 8\sqrt{3}$

Q. $x^2+\frac{1}{x^2}=4$ $\qquad$ $x^4-\frac{1}{x^4}=?$

Soln $\quad$ let $x^2=p$

$$p+\frac{1}{p}=4$$

$$p^2-\frac{1}{p^2}=\left(p-\frac{1}{p}\right)\left(p+\frac{1}{p}\right)$$

$$=\sqrt{4^2-4}\times 4$$

$$=2\sqrt{3}\times 4$$

$$=8\sqrt{3}$$

Q. $x^2+\frac{1}{x^2}=4$

$$x^4-\frac{1}{x^4}=\left(x^2+\frac{1}{x^2}\right)\left(x^2-\frac{1}{x^2}\right)$$

$$= 4 \times \sqrt{4^2-4}$$

$$= 4 \times 2\sqrt{3}$$

$$= 8\sqrt{3}$$

Q. $x^3+\frac{1}{x^3}=3$

Ans $x^6-\frac{1}{x^6}=\left(x^3+\frac{1}{x^3}\right)\left(x^3-\frac{1}{x^3}\right)$

$$= 3 \times \sqrt{3^2-4}$$

$$= 3 \times \sqrt{5}$$

$$= \underline{3\sqrt{5}}$$

Q $x^4 + \frac{1}{x^4} = 3$

$$x^8 - \frac{1}{x^8} = \left(x^4 + \frac{1}{x^4}\right)\left(x^4 - \frac{1}{x^4}\right)$$

$$= 3 \times \sqrt{3^2 - 4}$$

$$= \underline{3\sqrt{5}}$$

23. If $\left(x^2+\frac{1}{x^2}\right)=6$ and $0<x<1$, what is the value of $x^4-\frac{1}{x^4}$?

SSC CGL 26/07/2023 (Shift-01)

(a) $24\sqrt{2}$ (b) $-24\sqrt{2}$

(c) $-12\sqrt{10}$ (d) $12\sqrt{10}$

$x^4-\frac{1}{x^4}=\left(x^2+\frac{1}{x^2}\right)\left(x^2-\frac{1}{x^2}\right)$

$=6\sqrt{6^2-4}$

$=6\sqrt{32}$

$=6\times4\sqrt{2}=24\sqrt{2}$

$x^2 = 9$

$\Rightarrow x = \pm\sqrt{9}$

$\Rightarrow \boxed{x = \pm 3}$

$x = \sqrt{9}$

$\boxed{x = 3}$

$$x^2 = 9$$

$$\Rightarrow x = \pm\sqrt{9}$$

$$\Rightarrow \boxed{x = \pm 3}$$

$x = 3$ or $x = -3$

* If $x + \frac{1}{x} = n$ $\quad x - \frac{1}{x} = \pm\sqrt{n^2 - 4}$

* If $x - \frac{1}{x} = n$ $\quad x + \frac{1}{x} = \pm\sqrt{n^2 + 4}$

$x^2 - \frac{1}{x^2} = \pm\sqrt{n^2-4}$

$= \pm\sqrt{32} = \pm 4\sqrt{2}$

-ve

~~$+4\sqrt{2}$~~ $-4\sqrt{2}$ ✓

23. If $\left(x^2 + \frac{1}{x^2}\right) = 6$ and $0 < x < 1$, what is the value of $x^4 - \frac{1}{x^4}$?

SSC CGL 26/07/2023 (Shift-01)

(a) $24\sqrt{2}$ (b) $-24\sqrt{2}$ ✓

(c) $-12\sqrt{10}$ (d) $12\sqrt{10}$

$\therefore x^4 - \frac{1}{x^4} = \left(x^2 + \frac{1}{x^2}\right)\left(x^2 - \frac{1}{x^2}\right)$

$= 6 \times (-4\sqrt{2})$

$= -24\sqrt{2}$

$0 < x < 1$

let $x = \frac{1}{2}$

23. If $\left(x^2+\frac{1}{x^2}\right)=6$ and $0<x<1$, what is the value of $\left(x^4-\frac{1}{x^4}\right)$?

$=\left(x^2+\frac{1}{x^2}\right)\left(x^2-\frac{1}{x^2}\right)$

SSC CGL 26/07/2023 (Shift-01)

(a) $24\sqrt{2}$ (b) $-24\sqrt{2}$

(c) $-12\sqrt{10}$ (d) $12\sqrt{10}$

ans $=\left(x^2+\frac{1}{x^2}\right)\left(x^2-\frac{1}{x^2}\right)$

$=\left(\frac{1}{4}+4\right)\left(\frac{1}{4}-4\right)$

$=+\frac{17}{4}\left(-\frac{17}{4}\right)$

$0<x<1$



0.5

$x=0.5=\frac{1}{2}$

-ve

$x^2 - \frac{1}{x^2} = \pm\sqrt{n^2-4}$

$= \pm\sqrt{32} = \pm 4\sqrt{2}$

~~$4\sqrt{2}$~~ $-4\sqrt{2}$ ✓

23. If $\left(x^2 + \frac{1}{x^2}\right) = 6$ and $0 < x < 1$, what is the value of $\left(x^4 - \frac{1}{x^4}\right)$?

$= \left(x^2+\frac{1}{x^2}\right)\left(x^2-\frac{1}{x^2}\right)$

SSC CGL 26/07/2023 (Shift-01)

(a) $24\sqrt{2}$

(b) $-24\sqrt{2}$ ✓

(c) $-12\sqrt{10}$

(d) $12\sqrt{10}$

$\therefore$ ans $= \left(x^2+\frac{1}{x^2}\right)\left(x^2-\frac{1}{x^2}\right)$

$= 6 \times (-4\sqrt{2})$

$= -24\sqrt{2}$

$$x^4-\frac{1}{x^4}=\left(x^2+\frac{1}{x^2}\right)\left(x^2-\frac{1}{x^2}\right)$$

$$=\sqrt{(4\sqrt{2})^2+4}\times 4\sqrt{2}$$

$$=6\times 4\sqrt{2}=24\sqrt{2}$$

24. If $x^2-\frac{1}{x^2}=4\sqrt{2}$, what is the value of $x^4-\frac{1}{x^4}$?

SSC CGL 27/07/2023 (Shift-02)

(a) $16\sqrt{2}$

(b) $8\sqrt{2}$

(c) $24\sqrt{2}$

(d) $32\sqrt{2}$

25. If $\left(x+\frac{1}{x}\right)=5\sqrt{2}$, and $x>1$, what is the value of $\left(x^6-\frac{1}{x^6}\right)$?

SSC CHSL 02/08/2023 (Shift-02)

(a) $22970\sqrt{23}$ (b) $23030\sqrt{23}$

(c) $23060\sqrt{23}$ (d) $22960\sqrt{23}$

$x^3+\frac{1}{x^3}=(5\sqrt{2})^3-3\times5\sqrt{2}$

$=250\sqrt{2}-15\sqrt{2}$

$=235\sqrt{2}$

$ans=\left(x^3+\frac{1}{x^3}\right)\left(x^3-\frac{1}{x^3}\right)$

$=235\sqrt{2}\times\sqrt{(235\sqrt{2})^2-4}$

$=235\sqrt{2}\times\sqrt{55225\times2-4}$

$=235\sqrt{2}\times\sqrt{110446}$

$=235\sqrt{2}\times49\times\sqrt{2}\times\sqrt{23}$

$=470\times49\sqrt{23}$

$=23030\sqrt{23}$

$\sqrt{110446}=\sqrt{46\times2401}$

$=\sqrt{23\times2\times2401}$

25. If $\left(x+\frac{1}{x}\right)=5\sqrt{2}$, and $x>1$, what is the value of $\left(x^6-\frac{1}{x^6}\right)$?

SSC CHSL 02/08/2023 (Shift-02)

(a) $22970\sqrt{23}$ (b) $23030\sqrt{23}$

(c) $23060\sqrt{23}$ (d) $22960\sqrt{23}$

$x+\frac{1}{x}=5\sqrt{2}$

$x-\frac{1}{x}=\sqrt{50-4}=\sqrt{46}$

$x^3+\frac{1}{x^3}=(5\sqrt{2})^3-3\times5\sqrt{2}$

$=250\sqrt{2}-15\sqrt{2}$

$=235\sqrt{2}$

$x^3-\frac{1}{x^3}=(\sqrt{46})^3+3\sqrt{46}$

$=46\sqrt{46}+3\sqrt{46}$

$=49\sqrt{46}$

$=\left(x^3+\frac{1}{x^3}\right)\left(x^3-\frac{1}{x^3}\right)$

$ans=235\sqrt{2}\times49\times\sqrt{2}\times\sqrt{23}$

$=470\times49\times\sqrt{23}$

$=23030\sqrt{23}$

26. If $\left(x-\frac{1}{x}\right)=2\sqrt{2}$, and $x>1$, what is the value of $\left(x^6-\frac{1}{x^6}\right)$.

SSC CGL 20/07/2023 (Shift-03)

(a) $372\sqrt{6}$ (b) $384\sqrt{6}$

(c) $396\sqrt{6}$ (d) $420\sqrt{6}$

$x-\frac{1}{x}=2\sqrt{2}$

$x+\frac{1}{x}=\sqrt{(2\sqrt{2})^2+4}$

$=\sqrt{12}=2\sqrt{3}$

$x^3-\frac{1}{x^3}=(2\sqrt{2})^3+6\sqrt{2}$

$=16\sqrt{2}+6\sqrt{2}$

$=22\sqrt{2}$

$x^3+\frac{1}{x^3}=(2\sqrt{3})^3-6\sqrt{3}$

$=18\sqrt{3}$

$\left(x^3-\frac{1}{x^3}\right)\left(x^3+\frac{1}{x^3}\right)$

Ans $=22\sqrt{2}\times 18\sqrt{3}$

$=396\sqrt{6}$

27. If $\left(x+\frac{1}{x}\right)=5$, and $x>1$, what is the value of $\left(x^8-\frac{1}{x^8}\right)$?

SSC CGL 18/07/2023 (Shift-03)

(a) $60605\sqrt{21}$ (b) $60615\sqrt{21}$

(c) $60705\sqrt{21}$ (d) $60725\sqrt{21}$

$$x^8-\frac{1}{x^8}=\left(x^4+\frac{1}{x^4}\right)\left(x^2+\frac{1}{x^2}\right)\left(x+\frac{1}{x}\right)\left(x-\frac{1}{x}\right)$$

$$= 527 \times 23 \times 5 \times \sqrt{5^2-4}$$

$$= 527 \times 115 \times \sqrt{21}$$

$$= 60605\sqrt{21}$$

28. If $x + \frac{1}{x} = 15$, then the value of $\frac{7x^2 - 9x + 7}{x^2 - x + 1}$ is:

$$\frac{7x - 9 + \frac{7}{x}}{x - 1 + \frac{1}{x}} = \frac{7\left(x + \frac{1}{x}\right) - 9}{x + \frac{1}{x} - 1}$$

$$= \frac{7 \times 15 - 9}{15 - 1} = \frac{96}{14} = \frac{48}{7}$$

SSC CGL 12/09/2024 (Shift-01)

(a) $-\frac{48}{7}$

(b) $-\frac{22}{7}$

(c) $\frac{48}{7}$ ✓

(d) $\frac{22}{7}$

$$\frac{7}{x+\frac{1}{x}-8}$$

$$= \frac{7}{10-8} = \frac{7}{2} = 3.5$$

29. If $x+\frac{1}{x}=10$, then find the value of

$$\frac{7x}{x^2+1-8x}$$

.

SSC CHSL 03/06/2022 (Shift- 01)

(a) 3.5
(b) 4.5
(c) 2.5
(d) 5.5

$$\frac{x^3+\frac{1}{x^3}}{x-3+\frac{1}{x}} = \frac{110}{5-3}$$

$$= \frac{110}{2} \; 55$$

30. If $x+\frac{1}{x} = 5,\ x \neq 0$ then the value of $\frac{x^3+\frac{1}{x^3}}{x^2-3x+1}$ is equal to :

SSC CHSL 20/10/2020 (Shift- 01)

(a) 55 ✓

(b) 60

(c) 65

(d) 50

$x+\frac{1}{x}=n$

$x^3+\frac{1}{x^3}=n^3-3n$

$=125-15$

$=110$

T-1

$x + \frac{1}{x} = 3$

$x^2 + \frac{1}{x^2} = ?$

Soln

$x^2 + \frac{1}{x^2} = 3^2 - 2$

$= 7$

T-2

$x + \frac{1}{x} = 3$

$\frac{x^2+1+5x}{x^2+1+7x} = ?$

$\frac{\frac{x^2}{x} + \frac{1}{x} + \frac{5x}{x}}{\frac{x^2}{x} + \frac{1}{x} + \frac{7x}{x}}$

$\frac{x + \frac{1}{x} + 5}{x + \frac{1}{x} + 7} = \frac{3+5}{3+7} = \frac{8}{10} \; \frac{4}{5}$

T-3

$x^2 + 1 - 3x = 0$

$x^2 + \frac{1}{x^2} = ?$

$\frac{x^2}{x} + \frac{1}{x} - \frac{3x}{x} = \frac{0}{x}$

$\Rightarrow x + \frac{1}{x} - 3 = 0$

$\therefore x + \frac{1}{x} = 3$

$x^2 + \frac{1}{x^2} = 3^2 - 2$

$= 7$

T-4

$x^2 + 1 - 3x = 0$

$\frac{x^2+1+5x}{x^2+1+7x} = ?$

Sol^{n} $\quad \frac{x^2}{x}+\frac{1}{x}-\frac{3x}{x}=\frac{0}{x}$

$$x+\frac{1}{x}-3=0$$

$$\boxed{x+\frac{1}{x}=3}$$

$$\therefore \frac{x^2+1+5x}{x^2+1+7x}=\frac{x+\frac{1}{x}+5}{x+\frac{1}{x}+7}$$

$$=\frac{3+5}{3+7}=\frac{8}{10}\ \frac{4}{5}$$

T-4

$$x^2+1-3x=0$$

$$\frac{x^2+1+5x}{x^2+1+7x}=?$$

31. If $x - \frac{1}{x} = 5$, $x \neq 0$, then what is the value of $\frac{x^6 + 3x^3 - 1}{x^6 - 8x^3 - 1}$?

SSC CGL TIER II 16/11/2020

(a) $\frac{13}{12}$ (b) $\frac{11}{13}$

(c) $\frac{3}{8}$ (d) $\frac{4}{9}$

$$\frac{x^3 + 3 - \frac{1}{x^3}}{x^3 - 8 - \frac{1}{x^3}} = \frac{x^3 - \frac{1}{x^3} + 3}{x^3 - \frac{1}{x^3} - 8}$$

$$= \frac{143}{132} \quad \frac{13}{12}$$

$x - \frac{1}{x} = n$

$x^3 - \frac{1}{x^3} = n^3 + 3n$

$= 125 + 15$

$= 140$

32. If $x^2 + 3x + 1 = 0$, then what is the value of $x^6 + \frac{1}{x^6}$?

SSC CGL 3/03/2020 (Shift- 02)

(a) 324
(b) 322
(c) 318
(d) 327

$x + \frac{1}{x} = -3$

$x^6 + \frac{1}{x^6} = (n^3 - 3n)^2 - 2$

$= (-24 + 9)^2 - 2$

$= (-18)^2 - 2$

$= 324 - 2$

$= 322$

33. If $x^2 - 2\sqrt{5}x + 1 = 0$, then what is the value of $x^5 + \frac{1}{x^5}$?

SSC CGL 5/03/2020 (Shift- 01)

(a) $610\sqrt{5}$ ✓
(b) $408\sqrt{5}$
(c) $612\sqrt{5}$
(d) $406\sqrt{5}$

$x + \frac{1}{x} = 2\sqrt{5}$

$x^5 + \frac{1}{x^5} = (n^2-2)(n^3-3n) - n$

$= (20-2)(40\sqrt{5} - 6\sqrt{5}) - 2\sqrt{5}$

$= 18 \times 34\sqrt{5} - 2\sqrt{5}$

$= 612\sqrt{5} - 2\sqrt{5}$

$= 610\sqrt{5}$

$(2\sqrt{5})^3 = 8 \times 5\sqrt{5}$

$$\frac{1}{x-2+\frac{1}{x}} = \frac{1}{3}$$

$$\Rightarrow \quad 3 = x+\frac{1}{x}-2$$

$$\Rightarrow \quad 5 = x+\frac{1}{x}$$

$$x^3+\frac{1}{x^3} = 5^3-3\times 5$$

$$= 110$$

34. If $\dfrac{\frac{x}{x}}{\frac{x^2}{x}-\frac{2x}{x}+\frac{1}{x}} = \dfrac{1}{3}$, then the value of $x^3+\dfrac{1}{x^3}$ is:

SSC CGL 26/09/2024 (Shift-03)

(a) 110
(b) 130
(c) 160
(d) 180

$x - 5 + \frac{1}{x} = 0$

$\Rightarrow x + \frac{1}{x} = 5$

35. If $\frac{x^2}{x} - \frac{5x}{x} + \frac{1}{x} = 0$, then the value of $\frac{x^6}{x^3} + \frac{x^4}{x^3} + \frac{x^2}{x^3} + \frac{1}{x^3} = ?$

$5x^3$

x^3

SSC CGL 01/12/2022 (Shift-01)

(a) 30 (b) 25

(c) 23 (d) 28

$$\frac{x^3 + x + \frac{1}{x} + \frac{1}{x^3}}{5} = \frac{x^3 + \frac{1}{x^3} + x + \frac{1}{x}}{5}$$

$$= \frac{110 + 5}{5} = \frac{115}{5} = 23$$ Ans

$x+\frac{1}{x}=3$

$$\frac{x^3+\frac{1}{x^3}}{x+\frac{1}{x}+5}=\frac{18}{8}\ \frac{9}{4}$$

36. If $x^2 - 3x + 1 = 0$, then the value of

$$\frac{\frac{x^4}{x}+\frac{1}{x^2\cdot x^1}}{\frac{x^2}{x^1}+\frac{5x}{x}+\frac{1}{x}}$$ **is:**

SSC CGL MAINS 03/02/2022

(a) $\frac{9}{4}$

(b) $\frac{27}{8}$

(c) $\frac{5}{2}$

(d) 2

$x+\frac{1}{x}=3$

$x^3+\frac{1}{x^3}=3^3-3\times3=18$

37. If for a non-zero x, $7x^2 + 5x + 7 = 0$, then the $x^3 + \frac{1}{x^3}$ is ____.

SSC Phase-XII 24/06/2024 (Shift-02)

(a) $\frac{420}{343}$ (b) $\frac{610}{420}$

(c) $\frac{610}{343}$ (d) $\frac{343}{610}$

$7x + \frac{7}{x} = -5$

$\Rightarrow 7\left(x + \frac{1}{x}\right) = -5$

$\Rightarrow x + \frac{1}{x} = -\frac{5}{7}$

$x^3 + \frac{1}{x^3} = m^3 - 3m$

$= -\frac{125}{343} + \frac{15}{7}$

$= \frac{-125 + 735}{343} = \frac{610}{343}$

$x + \frac{8}{x} = -4$

$\frac{4}{x - 5 + \frac{8}{x}} = \frac{4}{-4-5} = \frac{4}{-12} \frac{1}{3}$

38. If $x^2 + 7x + 8 = 0$, then find the value of

$$\frac{4x}{x^2 - 5x + 8}$$

$\frac{4x}{x} \quad \frac{x}{x} \quad \frac{8}{x}$

SSC CGL 25/09/2024 (Shift-01)

(a) $\frac{1}{2}$

(b) $\frac{-1}{4}$

(c) $\frac{-1}{3}$ ✓

(d) $\frac{1}{6}$

* $x^{-1} = \frac{1}{x^1}$

* $x^{-2} = \frac{1}{x^2}$

* $x^{-3} = \frac{1}{x^3}$

* $\frac{1}{x^{-1}} = x^1$

* $\frac{1}{x^{-2}} = x^2$

* $\frac{1}{x^{-3}} = x^3$

* $(x+y)^{-1} = \frac{1}{(x+y)}$

* $(x+y)^{-2} = \frac{1}{(x+y)^2}$

* $(x+y+z)^{-1} = \frac{1}{(x+y+z)^1}$

$x - \frac{1}{x} = 3$

$$\frac{\left(x - \frac{1}{x} + 8\right)}{\left(x^2 + \frac{1}{x^2}\right)} = \frac{3+8}{3^2+2} = \frac{11}{11} = 1$$

39. If $x^2 - 3x - 1 = 0$, then the value of $(x^2 + 8x - 1)(x^3 + x^{-1})^{-1}$ is :

SSC CGL 07/06/2019 (Shift- 02)

(a) $\frac{3}{8}$ (b) 8

(c) 1 (d) 3

$$\left(\frac{x^2}{x} + \frac{8x}{x} - \frac{1}{x}\right) \times \frac{1}{\left(\frac{x^3}{x} + \frac{1}{x \cdot x}\right)}$$

Equal v/s Unequal

* $\left(x+\frac{1}{x}\right)^2=(3)^2$

$x^2+\frac{1}{x^2}+2\times x\times\frac{1}{x}=9$

$\Rightarrow x^2+\frac{1}{x^2}=9-2=7$

* $\left(2x+\frac{1}{2x}\right)^2=(3)^2$

$4x^2+\frac{1}{4x^2}+2\times 2x\times\frac{1}{2x}=9$

$4x^2+\frac{1}{4x^2}=7$

* $\left(3x+\frac{1}{2x}\right)^2=(6)^2$

$9x^2+\frac{1}{4x^2}+2\times 3x\times\frac{1}{2x}=36$

$9x^2+\frac{1}{4x^2}=33$

* $\left(4x+\frac{1}{6x}\right)^2=(30)^2$

$16x^2+\frac{1}{36x^2}+2\times 4x\times\frac{1}{6x}=900$

$16x^2+\frac{1}{36x^2}=900-\frac{4}{3}=\frac{2696}{3}$

Equal v/s Unequal

* $x+\frac{1}{x}=3$

$x^2+\frac{1}{x^2}=3^2-2=7$

* $2x+\frac{1}{2x}=3$

$4x^2+\frac{1}{4x^2}=3^2-2=7$

* $3x+\frac{1}{2x}=6$

$9x^2+\frac{1}{4x^2}=6^2-\cancel{2}\times3\times\frac{1}{\cancel{2}}=33$

* $4x^1+\frac{1}{6x}=30$

$16x^2+\frac{1}{36x^2}=(30)^2-\cancel{2}\times4\times\frac{1}{\cancel{6}3}$

$=900-\frac{4}{3}$

$=\frac{2696}{3}$

$3x + \frac{1}{3x} = 9$

40. If $3x + \frac{1}{9x} = 3 \times 3$ then the value of $9x^2 + \frac{1}{9x^2}$ is:

$9x^2 + \frac{1}{9x^2} = 9^2 - 2 = 79$

SSC CGL 11/09/2024 (Shift-03)

(a) 87
(b) 81
(c) 77
(d) 79

40. If $x + \frac{1}{9x} = 3$, then the value of $9x^2 + \frac{1}{9x^2}$ is:

SSC CGL 11/09/2024 (Shift-03)

(a) 87
(b) 81
(c) 77
(d) 79

Q. $4x + \frac{1}{2x} = 6$

$16x^2 + \frac{1}{4x^2} = 6^2 - 2 \times 4 \times \frac{1}{2}$

$= 32$

Q. $4x + \frac{1}{2x} = 3$

$49x^2 + \frac{1}{4x^2} = 3^2 - 2 \times 7 \times \frac{1}{2}$

$= 2$

Q. $2x + \frac{1}{3x} = 5$

(a) $4x^2 + \frac{1}{9x^2} =$

(b) $9x^2 + \frac{1}{4x^2}$

Soln

$4x^2 + \frac{1}{9x^2} = 5^2 - 2 \times 2 \times \frac{1}{3}$

$= 25 - \frac{4}{3}$

$= \frac{71}{3}$

* $2x + \frac{1}{3x} = 5$

$\times \frac{3}{2}$

$3x + \frac{1}{2x} = 5 \times \frac{3}{2} = \frac{15}{2}$

$9x^2 + \frac{1}{4x^2} = \left(\frac{15}{2}\right)^2 - 2 \times 3 \times \frac{1}{2}$

$= \frac{225}{4} - 3$

$= \frac{213}{4}$

$$* \quad 3x + \frac{1}{4x} = 15$$

$$16x^2 + \frac{1}{9x^2} = ?$$

Ans

$$3x + \frac{1}{4x} = 15$$

$$4x + \frac{1}{3x} = \frac{15}{3} \times 4 = 20$$

$$16x^2 + \frac{1}{9x^2} = (20)^2 - 2 \times 4 \times \frac{1}{3}$$

$$= 400 - \frac{8}{3}$$

$$= \frac{1192}{3}$$

41. If $5x + \frac{1}{3x} = 5$, find the value of $9x^2 + \frac{1}{25x^2}$

(a) 0 (b) 5

(c) 4 (d) 7.8

$5x + \frac{1}{3x} = 5$

$3x + \frac{1}{5x} = \frac{5}{5} \times 3 \times \frac{1}{5}$

$\therefore 9x^2 + \frac{1}{25x^2} = 3^2 - 2 \times 3 \times \frac{1}{5}$

$= 9 - 1.2$

$= 7.8$

* $x + \frac{1}{x} = 5$

$x^3 + \frac{1}{x^3} = n^3 - 3n$

$= 125 - 15$

$= \underline{110}$

* $x + \frac{1}{2x} = 5$

$x^3 + \frac{1}{8x^3} = n^3 - 3n \times 1 \times \frac{1}{2}$

$= 125 - 15 \times \frac{1}{2}$

$= \frac{250 - 15}{2} = \frac{235}{2}$

$3x + \frac{1}{2x} = 5$

$2x + \frac{1}{3x} = 5 \times 2 \times \frac{1}{3} = \frac{10}{3}$

42. If $3x + \frac{1}{2x} = 5$ then find $8x^3 + \frac{1}{27x^3}$

(a) $118\frac{1}{2}$ (b) $30\frac{10}{27}$

(c) 0 (d) 1

$\therefore \quad 8x^3 + \frac{1}{27x^3} = n^3 - 3n \times 2 \times \frac{1}{3}$

$= \frac{1000}{24} - 2 \times \frac{10}{3}$

$= \frac{1000 - 180}{24} = \frac{820}{24} = 30\frac{10}{24}$

Q. $2x + \frac{1}{3x} = 6$

$27x^3 + \frac{1}{8x^3} = ?$

Soln $2x + \frac{1}{3x} = 6$

$3x + \frac{1}{2x} = \frac{6}{2} \times 3x \frac{1}{2} = 9$

$27x^3 + \frac{1}{8x^3} = n^3 - 3n \times 3x \frac{1}{2}$

$= 729 - 3 \times 9 \times 3x \frac{1}{2}$

$= 729 - 40\frac{1}{2}$

$= 688\frac{1}{2}$ ✓

43. If $x + \frac{1}{16x} = 3$, then the value of $16x^3 + \frac{1}{256x^3}$ is :

SSC CGL TIER II 12/09/2019

(a) 423
(b) 441
(c) 432
(d) 414

$x + \frac{1}{16x} = 3$

$16\left(x^3 + \frac{1}{4096x^3}\right) = \left(n^3 - 3n \times 1 \times \frac{1}{16}\right)16$

$\Rightarrow \quad 16x^3 + \frac{1}{256x^3} = 16n^3 - 3n$

$= 16 \times 27 - 3 \times 3$

$= 432 - 9$

$= 423$

$5x+\frac{1}{3x}=4$

44. If $5x + \frac{1}{3x} = 4$, then the value of $9x^2 + \frac{1}{25x^2}$ is :

$3x+\frac{1}{5x}=4\times3\times\frac{1}{5}=\frac{12}{5}$

$9x^2+\frac{1}{25x^2}=\left(\frac{12}{5}\right)^2-2\times3\times\frac{1}{5}$

$=\frac{144}{25}-\frac{6}{5}\times\frac{5}{5}$

$=\frac{114}{25}$

SSC CGL 04/03/2020 (Shift- 01)

(a) $\frac{174}{125}$

(b) $\frac{144}{125}$

(c) $\frac{114}{25}$ ✓

(d) $\frac{119}{25}$

45. If $20x^2 - 30x + 1 = 0$, then what is the value of $25x^2 + \frac{1}{16x^2}$ is :

SSC CGL 05/03/2020 (Shift-02)

(a) $53\frac{1}{2}$ (b) $58\frac{1}{2}$

(c) $53\frac{3}{4}$ (d) $58\frac{3}{4}$

$5x - \frac{15}{2} + \frac{1}{4x} = 0$

$\Rightarrow 5x + \frac{1}{4x} = \frac{15}{2}$

$25x^2 + \frac{1}{16x^2} = \left(\frac{15}{2}\right)^2 - 2 \times 5 \times \frac{1}{4}$

$= \frac{225}{4} - \frac{10}{4}$

$= \frac{215}{4} = 53\frac{3}{4}$

$5x + \frac{1}{4x} =$

$4a + \frac{1}{5a} = 4$

$5a + \frac{1}{4a} = \frac{4}{4} \times 5 \times \frac{1}{4}$

$25a^2 + \frac{1}{16a^2} = 5^2 - 2 \times 5 \times \frac{1}{2 \cancel{4}}$

$= 25 - \frac{5}{2}$

$= \frac{45}{2}$

46. If $4a + \frac{1}{5a} = 4$, then the value of $25a^2 + \frac{1}{16a^2}$ is :

SSC CHSL 21/10/2020 (Shift-03)

(a) $\frac{45}{2}$ ✓

(b) $\frac{55}{2}$

(c) $\frac{43}{2}$

(d) $\frac{45}{4}$

* $x+\frac{1}{x}=5$

$x-\frac{1}{x}=\sqrt{5^2-4}$

$=\sqrt{21}$

* $2x+\frac{1}{3x}=6$

$2x-\frac{1}{3x}=\sqrt{n^2-4\times 2\times\frac{1}{3}}$

$=\sqrt{36-\frac{8}{3}}$

$=\sqrt{\frac{100}{3}}=\frac{10}{\sqrt{3}}$ ✓

47. If $2x^2 + 5x + 1 = 0$, then one of the values of $x - \frac{1}{2x}$?

SSC CGL MAINS 29/01/2022

(a) $\frac{\sqrt{17}}{2}$ (b) $\frac{13}{2}$

(c) $\frac{5}{2}$ (d) $\frac{\sqrt{13}}{2}$

$x + \frac{5}{2} + \frac{1}{2x} = 0$

$\therefore \quad x + \frac{1}{2x} = -\frac{5}{2}$

$x - \frac{1}{2x} = \sqrt{x^2 + \frac{1}{4x^2} - 2}$

$= \sqrt{\frac{25}{4} - 2}$

$= \sqrt{\frac{17}{4}} = \frac{\sqrt{17}}{2}$

$5x+\frac{1}{4x}=\sqrt{n^2+4\times5\times\frac{1}{4}}$

$=\sqrt{36+5}=\sqrt{41}$

48. If $5x-\frac{1}{4x}=6$, $x>0$, then find the value of $25x^2-\frac{1}{16x^2}$.

SSC CGL 21/04/2022 (Shift 01)

(a) $6\sqrt{41}$ (b) 36

(c) $\sqrt{246}$ (d) $6\sqrt{31}$

$(5x)^2-\frac{1}{(4x)^2}=\left(5x-\frac{1}{4x}\right)\left(5x+\frac{1}{4x}\right)$

$=6\times\sqrt{41}$

Basic

$4x^2+\frac{1}{4x^2}=2$

$2x+\frac{1}{2x}=\sqrt{2+2}=2$

$8x^3+\frac{1}{8x^3}=n^3-3n$

$=8-3\times2$

$=2$

49. If $\left(4x^2+\frac{1}{4x^2}\right)$ = 2,than what is the value of $\left(8x^3+\frac{1}{8x^3}\right)$?

(a) +2 (b) –2

(c) +4 (d) –4

$x^2+\frac{1}{x^2}=14$

$x+\frac{1}{x}=\sqrt{14+2}$

$4x^2 = 1$

$x^2 = \frac{1}{4}$

$\therefore x = \frac{1}{2}$

49. If $\left(4x^2 + \frac{1}{4x^2}\right) = 2$, than what is the value of $\left(8x^3 + \frac{1}{8x^3}\right)$?

(a) +2 (b) –2

(c) +4 (d) –4

$8x\frac{1}{8} + \frac{1}{8x\frac{1}{8}}$

$= 1 + 1$

$= 2$

$0.4x + \frac{1}{x} = 5$

50. If $\left(0.4x + \frac{1}{x}\right) = 5$, what is the value of $\left(0.064x^3 + \frac{1}{x^3}\right)$?

$0.064x^3 + \frac{1}{x^3} = m^3 - 3m \times 0.4x1$

$= 125 - 18 \times \frac{4}{10} \times 2$

$= 119$

SSC CGL 20/04/2022 (Shift 01)

(a) 119
(b) 125
(c) 110
(d) 105

Reverse Power

$$x^4 + \frac{1}{x^4} = 194$$

$$x^2 + \frac{1}{x^2} = \sqrt{194+2} = 14$$

$$x' + \frac{1}{x'} = \sqrt{14+2} = 4$$

$$* \; x^4 + \frac{1}{x^4} = 119$$

$$x^2 + \frac{1}{x^2} = \sqrt{119+2} = 11$$

$$x + \frac{1}{x} = \sqrt{11+2} = \underline{\sqrt{13}}$$

$16x^4 + \frac{1}{16x^4} = 14159$

51. If $16x^4 + \frac{1}{16x^4} = 14159$, then find the value of $2x + \frac{1}{2x}$.

$4x^2 + \frac{1}{4x^2} = \sqrt{14159+2} = 119$

SSC CPO 27/06/2024 (Shift-03)

(a) 9
(b) 11
(c) 12
(d) 15

$2x + \frac{1}{2x} = \sqrt{119+2} = 11$

$\sqrt{14161} = 119$

$x^4+\frac{1}{x^4} = 194$

$$x^2+\frac{1}{x^2} = \sqrt{194+2} = 14$$

$$x+\frac{1}{x} = \sqrt{14+2} = 4$$

$$x-\frac{1}{x} = \sqrt{14-2} = \sqrt{12} = 2\sqrt{3}$$

Q. $x^4+\frac{1}{x^4}=194$ $\quad x=?$

Ans

$$x^2+\frac{1}{x^2}=\sqrt{194+2}=14$$

$$x+\frac{1}{x}=\sqrt{14+2}=4 \qquad x-\frac{1}{x}=\sqrt{14-2}=2\sqrt{3}$$

$$\begin{array}{rl} & x+\frac{1}{x}=4 \\ + & x-\frac{1}{x}=2\sqrt{3} \\ \hline & 2x=4+2\sqrt{3} \end{array}$$

$$x=2+\sqrt{3}$$

Q. $x^{62} + \frac{1}{x^{62}} = 44$ $\quad x^{31} = ?$

Ans

$x^{31} + \frac{1}{x^{31}} = \sqrt{44+2} = 7$

$x^{31} - \frac{1}{x^{31}} = \sqrt{44-2} = 3\sqrt{5}$

$$x^{31} + \frac{1}{x^{31}} = 7$$
$$+ \quad x^{31} - \frac{1}{x^{31}} = 3\sqrt{5}$$
$$2 \cdot x^{31} = 7 + 3\sqrt{5}$$
$$\therefore x^{31} = \frac{7+3\sqrt{5}}{2}$$

52. If $x^2 + \frac{1}{x^2} = 27$, then find

(i) $x + \frac{1}{x}$ (ii) $x - \frac{1}{x}$

$x^2 + \frac{1}{x^2} = 27$

$x + \frac{1}{x} = \sqrt{29}$ $x - \frac{1}{x} = \sqrt{25} = 5$

53. If $x^4 + \frac{1}{x^4} = 23$, find

(i) $x + \frac{1}{x}$ (ii) $x^3 + \frac{1}{x^3}$

$x^4 + \frac{1}{x^4} = 23$

$x^2 + \frac{1}{x^2} = \sqrt{25} = 5$

$x + \frac{1}{x} = \sqrt{7}$

$x^3 + \frac{1}{x^3} = m^3 - 3m$

$= 7\sqrt{7} - 3\sqrt{7}$

$= 4\sqrt{7}$

$x^4 - 1442 + \frac{1}{x^4} = 0$

$\Rightarrow x^4 + \frac{1}{x^4} = 1442$

$x^2 + \frac{1}{x^2} = \sqrt{1442+2} = 38$

$x + \frac{1}{x} = \sqrt{38+2} = 2\sqrt{10}$

$x - \frac{1}{x} = \sqrt{38-2} = 6$

54. If $\frac{x^8}{x^4} - \frac{1442x^4}{x^4} + \frac{1}{x^4} = \frac{0}{x^4}$, then a possible value of $x - \frac{1}{x}$ is :

SSC CGL TIER-II (11/09/2019)

(a) 5
(b) 8
(c) 4
(d) 6

55. If $x > 0$ and $x^4 + \frac{1}{x^4} = 142$, what is the value of $x^7 + \frac{1}{x^7}$?

SSC CGL 20/07/2023 (Shift-02)

(a) $1561\sqrt{14}$ (b) $1563\sqrt{14}$

(c) $1560\sqrt{14}$ (d) $1562\sqrt{14}$

$x^4 + \frac{1}{x^4} = 142$

$x^2 + \frac{1}{x^2} = \sqrt{144} = 12$

$x + \frac{1}{x} = \sqrt{12+2} = \sqrt{14}$

$\therefore x^3 + \frac{1}{x^3} = (\sqrt{14})^3 - 3\times\sqrt{14}$

$= 14\sqrt{14} - 3\sqrt{14}$

$= \boxed{11\sqrt{14}}$

$x^7 + \frac{1}{x^7} = \left(x^3 + \frac{1}{x^3}\right)\left(x^4 + \frac{1}{x^4}\right) - \left(x + \frac{1}{x}\right)$

$= 11\sqrt{14} \times 142 - \sqrt{14}$

$= 1562\sqrt{14} - 1\sqrt{14}$

$= 1561\sqrt{14}$

56. If $x > 0$, and $x^4 + \frac{1}{x^4} = 2207$, what is the value of $x^7 + \frac{1}{x^7}$?

SSC CGL 02/12/2022 (Shift-01)

(a) 710649 (b) 710647
(c) 710654 (d) 710661

$x^4 + \frac{1}{x^4} = 2207$

$x^2 + \frac{1}{x^2} = \sqrt{2209} = 47$

$x + \frac{1}{x} = \sqrt{49} = 7$

$x^3 + \frac{1}{x^3} = 7^3 - 3 \times 7$

$= 343 - 21$

$= 322$

$x^7 + \frac{1}{x^7} = \left(x^3 + \frac{1}{x^3}\right)\left(x^4 + \frac{1}{x^4}\right) - \left(x + \frac{1}{x}\right)$

$= 322 \times 2207 - 7$

By unit digit

$= ___4 - 7$

$= ___7$

57. If $x^4 + x^{-4} = 47$, $(x > 0)$, then the value of $(2x - 3)^2$ is :

SSC CHSL 8/07/2019 (Shift-03)

(a) 2
(b) 3
(c) 5
(d) 4

$x^4 + \frac{1}{x^4} = 47$

$x^2 + \frac{1}{x^2} = \sqrt{49} = 7$

$x + \frac{1}{x} = \sqrt{9} = 3$

$\Rightarrow x + \frac{1}{x} = 3$

$\Rightarrow \frac{x^2+1}{x} = 3$

$\Rightarrow x^2 + 1 = 3x$

$\Rightarrow x^2 - 3x = -1$

$4x^2 + 9 - 2 \times 2x \times 3$

$= 4x^2 - 12x + 9$

$= 4(x^2 - 3x) + 9$

$= 4 \times (-1) + 9$

$= 5$

57. If $x^4 + x^{-4} = 47$, $(x > 0)$, then the value of $(2x - 3)^2$ is :

SSC CHSL 8/07/2019 (Shift-03)

(a) 2
(b) 3
(c) 5
(d) 4

$x^4 + \frac{1}{x^4} = 47$

$x^2 + \frac{1}{x^2} = \sqrt{49} = 7$

$x + \frac{1}{x} = \sqrt{9} = 3$

$x - \frac{1}{x} = \sqrt{3^2 - 4} = \sqrt{5}$

$2x = 3 + \sqrt{5}$

$(2x-3)^2$

$= (3 + \sqrt{5} - 3)^2$

$= 5$

58. If $x^4 + x^{-4} = 194$, $(x > 0)$, then the value of $(2x - 4)^2$ is :

SSC CHSL 09/07/2019 (Shift-01)

(a) 15
(b) 20
(c) 12
(d) 16

$x^4 + \frac{1}{x^4} = 194$

$x^2 + \frac{1}{x^2} = \sqrt{196} = 14$

$x + \frac{1}{x} = \sqrt{16} = 4$

$+ \quad x - \frac{1}{x} = \sqrt{4^2 - 4} = \sqrt{12} = 2\sqrt{3}$

$2x = 4 + 2\sqrt{3}$

$\therefore (2x-4)^2 = (4+2\sqrt{3}-4)^2 = 12$

$x^4+\frac{1}{x^4}=322$

$x^2+\frac{1}{x^2}=\pm\sqrt{324}=\pm18$

$x-\frac{1}{x}=\pm\sqrt{18-2}=\pm4$

59. If $x^4+\frac{1}{x^4}=322$, find $x^3-\frac{1}{x^3}$

(a) ±76 (b) 76

(c) -76 (d) 95

+4 -4

$x^3-\frac{1}{x^3}=n^3+3n$

$=64+12$

$=+76$

$x^3-\frac{1}{x^3}=n^3+3n$

$=-64-12$

$=-76$

Special Type

$x+\frac{1}{x}=2$

$x=1$

$x+\frac{1}{x}=-2$

$x=-1$

0. $x+\frac{1}{x}=2$

⊙ $x^2+\frac{1}{x^3}=1^2+\frac{1}{1^3}=1+1=2$

⊙ $x^{2024}+\frac{1}{x^{2025}}=1+\frac{1}{1}=2$

* $x+\frac{1}{x}=2$

⊙ $x^6+x^7+x^8=?$

$=1^6+1^7+1^8$

$=1+1+1$

$=3$

60. If $x + \frac{1}{x} = 2$, find

(i) $x^{20} + x^{19} + x^{18} + \ldots\ldots\ldots + x^2 + x + 1$

(ii) $x^{49} - x^{48} + x^{47} - x^{46} + \ldots\ldots + x^3 - x^2 + x + 1$

(iii) $x^{12} + \frac{1}{x^{12}}$

(iv) $x^{17} + \frac{1}{x^9}$

(v) $x^{15} - \frac{1}{x^{17}}$

60. **If $x + \frac{1}{x} = 2$, find**

$\therefore x = 1$

(i) $x^{20} + x^{19} + x^{18} + \ldots\ldots + x^2 + x^1 + 1$

$1 + 1 + 1 + \text{-----} + 1 + 1 + 1$

२० बार

$= 21$

$x = 1$

60. If $x + \frac{1}{x} = 2$, find

(ii) $x^{49} - x^{48} + x^{47} - x^{46} + \ldots\ldots + x^3 - x^2 + x + 1$

$1 - 1 + 1 - 1 + \ldots\ldots + 1 - 1 + 1 + 1$

$= 2$

60. If $x + \frac{1}{x} = 2$, find

$x = 1$

(iii) $x^{12} + \frac{1}{x^{12}}$

$= 1 + 1$

$= 2$

60. If $x + \frac{1}{x} = 2$, find

$x = 1$

(iv) $x^{17} + \frac{1}{x^9}$

$= 1 + 1$

$= 2$

60. If $x + \frac{1}{x} = 2$, find

$x = 1$

(v) $x^{15} - \frac{1}{x^{17}}$

$= 1 - 1$

$= 0$

$x = 1$

61. If $x + \frac{1}{x} = 2$, then find the value of $\left(x^4 + \frac{4}{x^4}\right)\left(x^3 - \frac{4}{x^3}\right)$.

SSC CPO 27/06/2024 (Shift-03)

(a) –15
(b) 15
(c) –10
(d) 10

$(1+4)(1-4)$
$= 5(-3)$
$= -15$

62. If $x + \frac{1}{x} = 2$, then find the value of

$$\left(x^{2723} + \frac{1}{x^{3356}}\right).$$

SSC CGL 18/09/2024 (Shift-02)

(a) 1
(b) 2
(c) 0
(d) −1

$x = 1$

$1 + 1$

$= 2$

$x + \frac{1}{x} = -2$

$x = -1$

⊙ $x^{1000} + \frac{1}{x^{1001}} = (-1)^{1000} + \frac{1}{(-1)^{1001}} = 1 - 1 = 0$

⊙ $x^8 + x^9$

$= (-1)^e + (-1)^o$

$= 1 - 1$

$= 0$

$(-1)^{odd} = -1$

$(-1)^{even} = 1$

63. If $x + \frac{1}{x} = -2$

(i) $x^{39} + x^{38} + \ldots\ldots\ldots + x^2 + x + 1$

(ii) $x^{100} - x^{99} + x^{98} - x^{97} + \ldots\ldots + x^2 - x + 1$

(iii) $x^5 - \frac{1}{x^5}$

(iv) $x^{18} + \frac{1}{x^{18}}$

(v) $x^7 - \frac{1}{x^9}$

63. If $x + \frac{1}{x} = -2$ $x = -1$

(i) $x^{39} + x^{38} + \ldots\ldots x^3 + x^2 + x + 1$

$= -1 + 1 + \ldots - 1 + 1 - 1 + 1$

$= 0$

63. If $x + \frac{1}{x} = -2$ $\quad x = -1$

(ii) $x^{100} - x^{99} + x^{98} - x^{97} + \ldots\ldots + x^2 - x^1 + 1$

$= 1 - (-1) + 1 - (-1) + \ldots\ldots + 1 - (-1) + 1$

$= \underbrace{1 + 1 + 1 + 1 + \ldots\ldots + 1 + 1 + 1}_{100 \text{ বার}}$

$= 101$

63. If $x + \frac{1}{x} = -2$

$x = -1$

(iii) $x^5 - \frac{1}{x^5}$

$-1 - \frac{1}{(-1)}$

$= -1 + 1$

$= 0$

$x = -1$

63. If $x + \frac{1}{x} = -2$

(iv) $x^{18} + \frac{1}{x^{18}}$

$= 1 + 1$

$= 2$

$x=-1$

63. If $x + \frac{1}{x} = -2$

(v) $x^7 - \frac{1}{x^9}$

$= (-1) - (-1)$

$= -1+1$

$= 0$

$x + \frac{1}{x} = 2$ $\quad x = 1$

$x + \frac{1}{x} = -2$ $\quad x = -1$

$x + \frac{1}{x} = 1$ $\quad x^3 = -1$

$x + \frac{1}{x} = -1$ $\quad x^3 = 1$

$x+\frac{1}{x}=1$

Proof:

$$x+\frac{1}{x}=1$$

$$\Rightarrow \frac{x^2+1}{x}=1$$

$$\Rightarrow x^2+1=x$$

$$\Rightarrow (x+1)(x^2+1-x)=0\times(x+1)$$

$$\Rightarrow x^3+1^3=0$$

$$\Rightarrow \boxed{x^3=-1}$$ ∎

$$(a^3+b^3)=(a+b)(a^2+b^2-ab)$$

$$(x^3+1^3)=(x+1)(x^2+1-x)$$

64. If $x + \frac{1}{x} = 1$, find

(i) $x^{18} + x^{12} + x^{6} + 1$

(ii) $x^{99} + x^{96} + x^{93} + x^{90} + \ldots\ldots + x^{6} + x^{3} + 1$

(iii) $x^{45} - x^{42} + x^{39} - x^{36} + \ldots\ldots + x^{9} - x^{6} + x^{3} + 1$

(iv) $x^{32} + x^{46} + x^{90} + 2$

64. If $x + \frac{1}{x} = 1$, find

$x^3 = -1$

(i) $x^{18} + x^{12} + x^6 + 1$

$= (x^3)^6 + (x^3)^4 + (x^3)^2 + 1$

$= (-1)^6 + (-1)^4 + (-1)^2 + 1$

$= 1 + 1 + 1 + 1$

$= 4$

Q. $x + \frac{1}{x} = 1$ $\quad x^3 = -1$

$x^{20} + x^{17} + x^7 + x^4$

Soln

$x^{14}(x^3+1) + x^4(x^3+1)$

$= x^{14}(-1+1) + x^4(-1+1)$

$= 0 + 0$

$= 0$

$x^{20} + x^{17} + x^7 + x^4$

Trick:

~~x^{20}~~ $+ x^{17}$ $+$ ~~x^7~~ $+$ ~~x^4~~

Power में 3 का gap को cancel out कर देंगे इस से।

Q $x + \frac{1}{x} = 1$ $x^3 = -1$

$x^{2000} + x^{1994} + x^{2004} + x^{2001} + x^6 + x^3$

$= 0$

64. If $x + \frac{1}{x} = 1$, find

(ii) $x^{99} + x^{96} + x^{93} + x^{90} + \ldots\ldots x^{9} + x^{6} + x^{3} + 1$

$-1+1$

$= 0$

1, 4, 7, 10, ... 91

$$n = \frac{91 - 1}{3} + 1$$

$$= \frac{90}{3} + 1$$

$$= 30 + 1$$

$$= 31 \text{ पद}$$

$$n = \frac{a_n - a}{d} + 1$$

$n = \frac{45-3}{3} + 1$

$= \frac{42}{3} + 1$

$= 15$ Ans

64. If $x + \frac{1}{x} = 1$, find $x^3 = -1$

(iii) $x^{45} - x^{42} + x^{39} - x^{36} + \ldots\ldots + x^9 - x^6 + x^3 + 1$

$= \underbrace{-1-1-1-1\ldots\ldots-1-1-1}_{\text{15 बार}} + 1$

$= -15 + 1$

$= -14$

Q $x + \frac{1}{x} = 1$ $x^3 = -1$

* $x^{46} - x^{43} + x^{40} - x^{34} + \dots - + x^{10} - x^7 + x^4 + 1$

$= x^1 \cdot x^{45} - x^1 \cdot x^{42} + x \cdot x^{39} - x \cdot x^{36} + \dots - + x \cdot x^9 - x \cdot x^6 + x \cdot x^3 + 1$

$= x \cdot (x^3)^{15} - x(x^3)^{14} + x \cdot (x^3)^{13} - x(x^3)^{12} + \dots - + x \cdot (x^3)^3 - x \cdot (x^3)^2 + x \cdot (x^3)^1 + 1$

$= x(-1) - x(1) + x(-1) - x(1) + \dots + x(-1) - x(1) + x(-1) + 1$

$= -x - x - x - x \dots - - - - - x - x - x + 1$

$= -15x + 1$

64. If $x + \frac{1}{x} = 1$, find $x^3 = -1$

(iv) $x^{32} + x^{46} + x^{90} + 2$

$= \frac{x^{33}}{x} + x^1 \cdot x^{45} + (x^3)^{30} + 2$

$= \frac{(x^3)^{11}}{x} + x \cdot (x^3)^{15} + (x^3)^{30} + 2$

$= \frac{-1}{x} - x + 1 + 2$

$= -\left(x + \frac{1}{x}\right) + 3$

$= -1 + 3$

$= 2$

$x+\frac{1}{x}=1$

$\Rightarrow \frac{x^2+1}{x}=1$

$\Rightarrow x^2+1=x$

$\Rightarrow x^2-x=-1$

64. If $x+\frac{1}{x}=1$, find

(iv) $x^{32}+x^{46}+x^{90}+2$

$=(x^3)^{10}\cdot x^2+(x^3)^{15}\cdot x^1+(x^3)^{30}+2$

$=1\times x^2-1\times x+1+2$

$=x^2-x+1+2$

$=-x+x+2$

$=2$

Q. $x + \frac{1}{x} = 1$

$x^3 = -1$

$$x^{31} + x^{29} + x^{16} + x^{14} + 5$$

Soln $x^{30} \cdot x + x^{29} \cdot \frac{x^1}{x} + x^1 \cdot x^{15} + x^{14} \cdot \frac{x^1}{x} + 5$

$= (-1)^{10} \cdot x + \frac{(-1)^{10}}{x} + x(-1)^5 + \frac{(-1)^5}{x} + 5$

$= \cancel{x} + \cancel{\frac{1}{x}} - \cancel{x} - \cancel{\frac{1}{x}} + 5$

Type-1

$x+\frac{1}{x}=2 \quad ; \quad x=1$

$x+\frac{1}{x}=-2 \quad ; \quad x=-1$

Type 2:

$x+\frac{1}{x}=1 \quad ; \quad x^3=-1$

$x+\frac{1}{x}=-1 \quad ; \quad x^3=1$

Type: 3

$x+\frac{1}{x}=\sqrt{3} \quad ; \quad x^6=-1$

$x+\frac{1}{x}=-\sqrt{3} \quad ; \quad x^6=-1$

$$x+\frac{1}{x}=\sqrt{3}$$

Proof:

$$x^3+\frac{1}{x^3}=(\sqrt{3})^3-3\times\sqrt{3}$$

$$\therefore\quad x^3+\frac{1}{x^3}=3\sqrt{3}-3\sqrt{3}$$

$$x^3+\frac{1}{x^3}=0$$

$$\Rightarrow x^3=-\frac{1}{x^3}$$

$$\Rightarrow \boxed{x^6=-1}$$

$$x+\frac{1}{x}=-\sqrt{3}$$

$$x^3+\frac{1}{x^3}=(-\sqrt{3})^3-3(-\sqrt{3})$$

$$x^3+\frac{1}{x^3}=-3\sqrt{3}+3\sqrt{3}$$

$$x^3+\frac{1}{x^3}=0$$

$$\Rightarrow x^3=-\frac{1}{x^3}$$

$$\Rightarrow \boxed{x^6=-1}$$

65. If $x + \frac{1}{x} = \sqrt{3}$, find

(i) $x^{90} + \frac{1}{x^{90}}$ (ii) $x^{92} + \frac{1}{x^{92}}$

(iii) $x^{102} + x^{96} + x^{101} + x^{95} + x^{100} + x^{94}$ is

(iv) $x^6 - \frac{1}{x^6} + 2$ (v) $x^{18} + x^{12} + x^6 + 1$

$x+\frac{1}{x}=\sqrt{3}$

$\therefore \boxed{x^6=-1}$

65. If $x+\frac{1}{x}=\sqrt{3}$, find

(i) $x^{90}+\frac{1}{x^{90}}$

$= (x^6)^{15}+\frac{1}{(x^6)^{15}}$

$= (-1)^{15}+\frac{1}{(-1)^{15}}$

$= -1-1$

$= -2$

$\equiv$

65. If $x + \frac{1}{x} = \sqrt{3}$, find

(ii) $x^{92} + \frac{1}{x^{92}}$

$= (x^6)^{15} \cdot x^2 + \frac{1}{(x^6)^{15} \cdot x^2}$

$= -x^2 - \frac{1}{x^2} = -\left(x^2 + \frac{1}{x^2}\right)$

$= -\{(\sqrt{3})^2 - 2\}$

$= -1$

$=$

65. If $x + \frac{1}{x} = \sqrt{3}$, find $x^6 = -1$

(iii) $x^{102} + x^{96} + x^{101} + x^{95} + x^{100} + x^{94}$ is

$= 0$

$x^6 = -1$

65. If $x + \frac{1}{x} = \sqrt{3}$, find

(iv) $x^6 - \frac{1}{x^6} + 2$

$= -1 - \frac{1}{(-1)} + 2$

$= -1 + 1 + 2$

$= 2$

65. If $x + \frac{1}{x} = \sqrt{3}$, find

(v) $x^{18} + x^{12} + x^6 + 1$

$= -1 + 1$

$= 0$

$x+\frac{1}{x}=\sqrt{3}$

$x^6+\frac{1}{x^6}$

$=-1-1$

$=-2$

66. If $\left(x+\frac{1}{x}\right)^2 = 3$, then what is the value of $x^6 + x^{-6}$?

SSC CGL MAINS (08/08/2022)

(a) 6
(b) 2
(c) –2
(d) –6

67. If $x + \frac{1}{x} = \sqrt{3}$, then the value of $x^{18} + x^{12} + x^6 + 1$ is :

SSC CHSL 21/10/2020 (Shift- 03)

(a) 0
(b) 2
(c) 3
(d) 1

$= -1+1$

$= 0$

$x^6 = -1$

68. If $x + \frac{1}{x} = -\sqrt{3}$ then $x^{67} + x^{53} + x^{43} + x^{29} + x^{24} + x^{12} + x^6 + 3$ is

(a) $\sqrt{3}$ (b) 0

(c) $2(2+\sqrt{3})$ (d) $2(2-\sqrt{3})$

$$x^{67} + x^{53} + x^{43} + x^{29} + x^{24} + x^{12} + x^6 + 3$$

$$= x \cdot x^{66} + \frac{x^{54}}{x} + x \cdot x^{42} + \frac{x^{30}}{x} + (x^6)^4 + 3$$

$$= x \cdot (-1)^{11} + \frac{(-1)^9}{x} + x \cdot (-1)^7 + \frac{(-1)^5}{x} + (-1)^4 + 3$$

$$= -x - \frac{1}{x} - x - \frac{1}{x} + 1 + 3$$

$$-\left(x + \frac{1}{x}\right) - \left(x + \frac{1}{x}\right) + 4$$

$$= -(-\sqrt{3}) - (-\sqrt{3}) + 4$$

$$= 2\sqrt{3} + 4$$

$$= 2(\sqrt{3} + 2)$$

M-1 (Basic)

69. If $x = 2 + \sqrt{3}$, find $x + \frac{1}{x}$

(a) 5 (b) 4

(c) −4 (d) $2\sqrt{3}$

$x = 2+\sqrt{3}$

$\frac{1}{x} = \frac{1}{(2+\sqrt{3})} \times \frac{(2-\sqrt{3})}{(2-\sqrt{3})}$

$= \frac{(2-\sqrt{3})}{2^2-(\sqrt{3})^2} = \frac{2-\sqrt{3}}{4-3}$

$= 2-\sqrt{3}$

$x + \frac{1}{x}$

$= 2+\sqrt{3}+2-\sqrt{3}$

$= 4$

$x - \frac{1}{x}$

$= (2+\sqrt{3}) - (2-\sqrt{3})$

$= 2+\sqrt{3}-2+\sqrt{3}$

$= 2\sqrt{3}$

$$x = 2+\sqrt{3}$$

$$\boxed{x = a+\sqrt{b}} \qquad a > b$$

Condition : If $a^2 - b^2 = 1$

$$x + \frac{1}{x} = 2a = 2 \times \text{Bigger no.}$$

$$x - \frac{1}{x} = 2\sqrt{b} = 2 \times \text{Smaller no.}$$

M-2 (Smart Approach)

69. If $x = 2 + \sqrt{3}$, find $x + \frac{1}{x}$

$x + \frac{1}{x} = 2 \times 2 = 4$

$x - \frac{1}{x} = 2 \times \sqrt{3} = 2\sqrt{3}$

(a) 5 (b) 4

(c) −4 (d) $2\sqrt{3}$

$2^2 - \sqrt{3}^2$
$= 4 - 3$
$= 1$ ✓

70. If $x = 7 + 4\sqrt{3}$ then $x - \frac{1}{x}$

$x + \frac{1}{x} = 2 \times 7 = 14$

$x - \frac{1}{x} = 2 \times 4\sqrt{3} = 8\sqrt{3}$

(a) $8\sqrt{3}$ (b) 4

(c) -4 (d) $-2\sqrt{3}$

$49 - 48$

$= 1$

71. If $x = 7 - 4\sqrt{3}$, then find the value of $\left(x^2 + \frac{1}{x^2}\right)$.

SSC CGL 18/09/2024 (Shift-01)

(a) 188
(b) 194
(c) 186
(d) 196

$x + \frac{1}{x} = 2 \times 7 = 14$

$x^2 + \frac{1}{x^2} = (14)^2 - 2$

$= 196 - 2 = 194$

$49 - 48$

$= 1$

$x - \frac{1}{x} = 2 \times 4\sqrt{3} = 8\sqrt{3}$

71. If $x = 7 - 4\sqrt{3}$, then find the value of $\left(x^2 + \frac{1}{x^2}\right)$.

$x - \frac{1}{x} = 2 \times (-4\sqrt{3}) = -8\sqrt{3}$

SSC CGL 18/09/2024 (Shift-01)

(a) 188
(b) 194
(c) 186
(d) 196

72. If $x = 3 + 2\sqrt{2}$, then the value of $\sqrt{x} - \frac{1}{\sqrt{x}}$ is:

SSC CHSL 26/10/2020 (Shift-03)

(a) 2
(b) 1
(c) 0
(d) 3

$\therefore \quad x + \frac{1}{x} = 2 \times 3 = 6$

$\sqrt{x} + \frac{1}{\sqrt{x}} = \sqrt{6+2} = 2\sqrt{2}$

$\sqrt{x} - \frac{1}{\sqrt{x}} = \sqrt{6-2} = 2$

M-1 (Basic)

73. If $x = 1 + \sqrt{2}$, then the value of $\sqrt{x} + \left(\frac{1}{\sqrt{x}}\right)$ is:

SSC CHSL 17/03/2020 (Shift-03)

(a) 2.1014
(b) 2.1973
(c) 1.9876
(d) 1.9996

$x = \sqrt{2}+1$

$\frac{1}{x} = \frac{1}{\sqrt{2}+1} = \frac{\sqrt{2}-1}{1}$

$\therefore x + \frac{1}{x} = \sqrt{2}+1+\sqrt{2}-1$

$= 2\sqrt{2}$

$\sqrt{x} + \frac{1}{\sqrt{x}} = \sqrt{2\sqrt{2}+2} = \sqrt{2(\sqrt{2}+1)}$

$= \sqrt{2 \times 2.414} = \sqrt{4.82} = 2.1973$

$\sqrt{2} = 1.414$

$\sqrt{4} = 2$

$\sqrt{4.84} = 2.2$

$x + \frac{1}{x} = 2\sqrt{2}$

$\therefore \sqrt{x} + \frac{1}{\sqrt{x}} = \sqrt{2\sqrt{2}+2}$

$= \sqrt{2.828+2}$

$= \sqrt{4.82} = 2.1973$

73. If $x = 1 + \sqrt{2}$, then the value of $\sqrt{x} + \left(\frac{1}{\sqrt{x}}\right)$ is:

$x = \sqrt{2} + 1$

SSC CHSL 17/03/2020 (Shift-03)

(a) 2.1014
(b) 2.1973
(c) 1.9876
(d) 1.9996

$49 - 48 = 1$

74. If $p = 7 + 4\sqrt{3}$, then what is the value of $\frac{p^6 + p^4 + p^2 + 1}{p^3}$?

SSC CGL 06/12/2022 (Shift-02)

(a) 2617 (b) 2167

(c) 2716 (d) 2176

$p + \frac{1}{p} = 14$

$\therefore p^3 + \frac{1}{p^3} = 14^3 - 3 \times 14$

$= 2744 - 42$

$= 2702$

$p^3 + p^1 + \frac{1}{p} + \frac{1}{p^3}$

$= 14 + 2702$

$= 2716$

Special Concept

$PQ = 1$

$P = \frac{1}{Q}$

$P = 7 + 4\sqrt{3}$ $\quad Q = 7 - 4\sqrt{3}$

$P \times Q = 1$

$P+\frac{1}{P}=14$

$P^2+\frac{1}{P^2}=14^2-2=194$

**

75. If $P = 7 + 4\sqrt{3}$ and $PQ = 1$, then what is the value of $\left(\frac{1}{P^2}+\frac{1}{Q^2}\right)$ = ?

(a) 148
~~(b) 189~~
(c) 194
~~(d) 204~~

$\left(\frac{1}{P^2}+P^2\right)$

$P \times Q = 1$
$\Rightarrow P = \frac{1}{Q}$
$P^2 = \frac{1}{Q^2}$

$$* \quad x = \frac{a-b}{a+b} \qquad y = \frac{a+b}{a-b}$$

$$\left.\begin{aligned} x+y &= \frac{2(a^2+b^2)}{a^2-b^2} \\ y-x &= \frac{2 \times 2ab}{a^2-b^2} \end{aligned}\right\}$$

76. If $x = \frac{\sqrt{5}-\sqrt{4}}{\sqrt{5}+\sqrt{4}}$ and $y = \frac{\sqrt{5}+\sqrt{4}}{\sqrt{5}-\sqrt{4}}$ then the value of $\frac{x^2-xy+y^2}{x^2+xy+y^2}$ = ?

SSC CGL 08/12/2022 (Shift-02)

(a) $\frac{361}{363}$ (b) $\frac{341}{343}$

(c) $\frac{384}{387}$ (d) $\frac{321}{323}$

$x+\frac{1}{x} = \frac{\sqrt{5}-\sqrt{4}}{\sqrt{5}+\sqrt{4}} + \frac{\sqrt{5}+\sqrt{4}}{\sqrt{5}-\sqrt{4}}$

$= \frac{(\sqrt{5}-\sqrt{4})^2+(\sqrt{5}+\sqrt{4})^2}{(\sqrt{5}+\sqrt{4})(\sqrt{5}-\sqrt{4})}$

$= \frac{5+4-2\sqrt{20}+5+4+2\sqrt{20}}{5-4}$

$x+\frac{1}{x} = 18$

$\therefore x^2+\frac{1}{x^2} = 18^2-2 = 322$

$= \frac{x^2+\frac{1}{x^2}-1}{x^2+\frac{1}{x^2}+1}$

ans $= \frac{322-1}{322+1} = \frac{321}{323}$

$x = \frac{1}{y}$ or $y = \frac{1}{x}$

$\Rightarrow xy = 1$

76. If $x = \frac{\sqrt{5}-\sqrt{4}}{\sqrt{5}+\sqrt{4}}$ and $\frac{1}{x} = \frac{\sqrt{5}+\sqrt{4}}{\sqrt{5}-\sqrt{4}}$ then the value of $\frac{x^2 - xy + y^2}{x^2 + xy + y^2} = ?$ =

SSC CGL 08/12/2022 (Shift-02)

(a) $\frac{361}{363}$

(b) $\frac{341}{343}$

(c) $\frac{384}{387}$

(d) $\frac{321}{323}$ ✓

$x + \frac{1}{x} = \frac{2(a^2+b^2)}{a^2-b^2}$

$= \frac{2(5+4)}{5-4} = \frac{18}{1}$

$\therefore x^2 + \frac{1}{x^2} = 18^2 - 2 = 322$

Ans $\frac{x^2 + \frac{1}{x^2} - 1}{x^2 + \frac{1}{x^2} + 1} = \frac{321}{323}$

5 4

Q. $x=\frac{\sqrt{7}-\sqrt{6}}{\sqrt{7}+\sqrt{6}}$ $y=\frac{\sqrt{7}+\sqrt{6}}{\sqrt{7}-\sqrt{6}}$ $\frac{x^2+y^2-xy}{x^2+y^2+xy}$

Soln $\boxed{x=\frac{1}{y} \text{ or } y=\frac{1}{x} \text{ or } xy=1}$

$\therefore x+\frac{1}{x}=2x\frac{(a^2+b^2)}{(a^2-b^2)}=\frac{26}{1}$

$\therefore x^2+\frac{1}{x^2}=26^2-2=674$

ans $\frac{x^2+\frac{1}{x^2}-1}{x^2+\frac{1}{x^2}+1}=\frac{673}{675}$

Q. $x=\frac{\sqrt{8}-\sqrt{7}}{\sqrt{8}+\sqrt{7}}$ $y=\frac{\sqrt{8}+\sqrt{7}}{\sqrt{8}-\sqrt{7}}$ $\frac{x^2+y^2-xy}{x^2+y^2+xy}$

Ans $x+\frac{1}{x}=\frac{2\times 15}{1}=30$

$x^2+\frac{1}{x^2}=30^2-2=898$

$\text{ans}=\frac{x^2+\frac{1}{x^2}-1}{x^2+\frac{1}{x^2}+1}=\frac{897}{899}$

$(-3+m)+\frac{1}{m-3}=5-3$

$\Rightarrow (m-3)+\frac{1}{(m-3)}=2$

1 1

77. If $m+\frac{1}{m-3}=5$, then $(m-3)^8+\frac{1}{(m-3)^8}=?$

(a) 2 (b) 0

(c) −2 (d) 15

$ans = (m-3)^8+\frac{1}{(m-3)^8}$

$= 1+1$

$= 2$

$x+\frac{1}{x}=2$

$x=1$

77. If $m + \frac{1}{m-3} = 5$, then $(m-3)^8 + \frac{1}{(m-3)^8} = ?$

(a) 2
(b) 0
(c) –2
(d) 15

$m - 3 = 1$

$1 + 1 = 2$

Q. $m + \frac{1}{m-4} = 9$

$(m-4)^{2002} + \frac{1}{(m-4)^{2003}}$

Ans $(m-4) + \frac{1}{(m-4)} = 9-4 = 2$

$(m-4)^{2002} + \frac{1}{(m-4)^{2003}}$

$= 1^{2002} + \frac{1}{1^{2003}}$

$= 1 + 1$

$= 2$

$(x-9)+\frac{1}{(x-9)} = 11-9=2$

If $x+\frac{1}{x-9} = 11$, then $(x-9)^{12} + \frac{1}{(x-9)^{13}} = ?$

ans $= (x-9)^{12}+\frac{1}{(x-9)^{13}}$

$= 1^{12}+\frac{1}{1^{13}}$

$= 2$

(a) 0
(b) 2
(c) –2
(d) 1

$(x-11) = -\frac{1}{(x-9)}$

$\Rightarrow (x-9)-2 = \frac{-1}{(x-11)} - 2$

$\Rightarrow (x-11) = \frac{-1}{(x-11)} - 2$

$\Rightarrow \underbrace{(x-11)}_{-1} + \underbrace{\frac{1}{(x-11)}}_{-1} = -2$

78. If $x + \frac{1}{x-9} = 11$, then $(x-11)^{12} + \frac{1}{(x-11)^{13}} = ?$

(a) 0 (b) 2

(c) −2 (d) 1

$(x-11)^{12} + \frac{1}{(x-11)^{13}}$

$= (-1)^{12} + \frac{1}{(-1)^{13}}$

$= 1-1$

$= 0$

Q. $m + \frac{1}{m-11} = 13$

$(m-11)^8 + \frac{1}{(m-11)^8}$

~~Ans~~

$(m-11) = 1$

$1^8 + \frac{1}{1^8} = 2$

Q. $m + \frac{1}{m-11} = 13$

$(m-13)^8 + \frac{1}{(m-13)^8}$

$(m-13) = -1$

$(-1)^8 + \frac{1}{(-1)^8}$

$= 1 + 1$

$= 2$

78. If $x+\frac{1}{x-9}=11$, then $(x-11)^{12}+\frac{1}{(x-11)^{13}}=?$

(a) 0
(b) 2
(c) −2
(d) 1

$x-11=-1$

$(x-11)^{12}+\frac{1}{(x-11)^{13}}$

$=1-1$

$=0$

Q. $m + \frac{1}{m-3} = 5$

$m - 3 = 1$

$\therefore \boxed{m = 4}$

(a) $(m-3)^2 + \frac{1}{(m-3)^3} = 1^2 + \frac{1}{1^3} = 2$

(b) $(m-5)^2 + \frac{1}{(m-5)^2} = (-1)^2 + \frac{1}{(-1)^2} = 1 + 1 = 2$

(c) $(m-4)^2 + \frac{1}{(m-6)^2} = 0 + \frac{1}{(-2)^2} = \frac{1}{4}$ ✓

$a-4=1$

$a=5$

79. If $a+\frac{1}{a-4}=6$, then $(a-3)^7+\frac{1}{(a-7)^3}=?$

(a) $63\frac{7}{8}$

(b) $255\frac{1}{8}$

(c) $127\frac{7}{8}$

(d) 216

$2^7+\frac{1}{(-2)^3}$

$=128-\frac{1}{8}$

$=127\frac{7}{8}$

M-1 (Basic)

80. If $x^2 - 22x + 111 = 0$, then what is the value of $(x-8)^2 - \frac{1}{(x-8)^2} = ?$

(a) $12\sqrt{10}$ (b) $8\sqrt{5}$
(c) $8\sqrt{3}$ (d) 18

let $x-8=t$

$\Rightarrow x = t+8$

$\therefore (t+8)^2 - 22(t+8) + 111 = 0$

$\Rightarrow t^2 + 64 + 16t - 22t - 176 + 111 = 0$

$\Rightarrow t^2 - 6t - 1 = 0$

$\Rightarrow t - 6 - \frac{1}{t} = 0$

$\Rightarrow t - \frac{1}{t} = 6$

$\therefore t^2 - \frac{1}{t^2} = \left(t - \frac{1}{t}\right)\left(t + \frac{1}{t}\right)$

$= 6 \times \sqrt{6^2+4}$

$= 6 \times 2\sqrt{10}$

$= 12\sqrt{10}$

80. If $x^2 - 22x + 111 = 0$, then what is the value of $(x-8)^2 - \frac{1}{(x-8)^2} = ?$

(a) $12\sqrt{10}$ (b) $8\sqrt{5}$
(c) $8\sqrt{3}$ (d) 18

$+8 \times 2 - 22 = -6$

$\therefore t - \frac{1}{t} = +6$

$\therefore t^2 - \frac{1}{t^2} = \left(t - \frac{1}{t}\right)\left(t + \frac{1}{t}\right)$

$= 6\sqrt{6^2+4}$

$= 6 \times 2\sqrt{10}$

80. If $x^2 - 22x + 111 = 0$, then what is the value of $(x-8)^2 - \frac{1}{(x-8)^2} = ?$

(a) $12\sqrt{10}$ (b) $8\sqrt{5}$
(c) $8\sqrt{3}$ (d) 18

$+8\times2-22=-6$

$t-\frac{1}{t}=6$

$\therefore t^2-\frac{1}{t^2}=\left(t-\frac{1}{t}\right)\left(t+\frac{1}{t}\right)$

$=6\times\sqrt{6^2+4}$

$=6\times2\sqrt{10}$

M-1 (Basic)

81. If $x^2 + 13x + 39 = 0$, then $(x+8)^5 - \frac{1}{(x+8)^5}$ = ?

(a) 393 ✓
(b) 396
(c) 392
(d) 394

$x + 8 = t$

$\Rightarrow \boxed{x = t - 8}$

$\therefore (t-8)^2 + 13(t-8) + 39 = 0$

$\Rightarrow t^2 + 64 - 16t + 13t - 104 + 39 = 0$

$\Rightarrow t^2 - 3t - 1 = 0$

$\Rightarrow t - 3 - \frac{1}{t} = 0$

$\therefore \boxed{t - \frac{1}{t} = 3}$

$t^5 - \frac{1}{t^5} = (n^2+2)(n^3+3n) - n$

$= 11 \times 36 - 3$

$= 396 - 3$

$= 393$

81. If $x^2 + 13x + 39 = 0$, then $(x+8)^5 - \frac{1}{(x+8)^5}$ = ?

(a) 393
(b) 396
(c) 392
(d) 394

$-8 \times 2 + 13 = -3$

$t - \frac{1}{t} = 3$

$\therefore t^5 - \frac{1}{t^5} = (n^2+2)(n^3+3n) - n$

$= 11 \times 36 - 3$

$= 396 - 3$

$= 393$

82. If $x^2 - 12x + 33 = 0$, then what is the value of $(x-4)^4 + \frac{1}{(x-4)^4} = ?$

(a) 227 (b) 326
(c) 167 (d) 194

$4 \times 2 - 12 = -4$

$t + \frac{1}{t} = 4$

$\therefore t^2 + \frac{1}{t^2} = 4^2 - 2 = 14$

$t^4 + \frac{1}{t^4} = 14^2 - 2 = 194$

$x^3 - \frac{1}{x^3} = n^3 + 3n$

$= 125 + 15 = 140$

83. If $(x - a)(x - b) = 1$ and $a - b + 5 = 0$, then $(x - a)^3 - \frac{1}{(x - a)^3} = ?$

(a) 110 (b) 1

(c) 125 (d) 140

$x(x - 5) = 1$

$\Rightarrow (x - 5) = \frac{1}{x}$

$\Rightarrow x - \frac{1}{x} = 5$

$b = 5$

Q. $x - \frac{1}{x} = 2$ $\quad x^5 - \frac{1}{x^5} = ?$

$x^7 - \frac{1}{x^7} = ?$

Ans $\left(x^5 - \frac{1}{x^5}\right) = \left(x^2 + \frac{1}{x^2}\right)\left(x^3 - \frac{1}{x^3}\right) - \left(x - \frac{1}{x}\right)$

$= x^5 - \frac{1}{x} + x - \frac{1}{x^5} + \frac{1}{x} - x$

Q. $x - \frac{1}{x} = 2$ $\quad x^5 - \frac{1}{x^5} = ?$

$x^7 - \frac{1}{x^7} = ?$

Ans $\left(x^5 - \frac{1}{x^5}\right) = \left(x^2 + \frac{1}{x^2}\right)\left(x^3 - \frac{1}{x^3}\right) - \left(x - \frac{1}{x}\right)$

$= (n^2+2)(n^3+3n) - n$

$= 6 \times 14 - 2$

$= \underline{82}$

Q. $x-\frac{1}{x}=2$ $\quad x^7-\frac{1}{x^7}=?$

$x-\frac{1}{x}=2$

$x^2+\frac{1}{x^2}=2^2+2=6$

$x^4+\frac{1}{x^4}=6^2-2$

$=34$

Ans $\left(x^7-\frac{1}{x^7}\right)=\left(x^3-\frac{1}{x^3}\right)\left(x^4+\frac{1}{x^4}\right)+\left(x-\frac{1}{x}\right)$

$x^7+\frac{1}{x}-x-\frac{1}{x^7}\quad +x-\frac{1}{x}$

Q. $x-\frac{1}{x}=2$ $\quad x^7-\frac{1}{x^7}=?$

$x-\frac{1}{x}=2$

$x^2+\frac{1}{x^2}=2^2+2=6$

$x^4+\frac{1}{x^4}=6^2-2$

$=34$

~~Ans~~ $\left(x^7-\frac{1}{x^7}\right)=\left(x^3-\frac{1}{x^3}\right)\left(x^4+\frac{1}{x^4}\right)+\left(x-\frac{1}{x}\right)$

$=(n^3+3n)\times 34+2$

$=(8+6)\times 34+2$

$=446+2$

$=\underline{478}$

84. If $x^5 - \frac{1}{x^5} = \sqrt{7}$, then find $x^{35} - \frac{1}{x^{35}} = ?$

(a) $791\sqrt{7}$ (b) $789\sqrt{7}$

(c) $790\sqrt{7}$ (d) $792\sqrt{7}$

let $x^5 = p$

$p - \frac{1}{p} = \sqrt{7}$

$\therefore p^2 + \frac{1}{p^2} = 7 + 2 = 9$

$p^4 + \frac{1}{p^4} = 9^2 - 2 = 79$

$p^7 - \frac{1}{p^7} = \left(p^3 - \frac{1}{p^3}\right)\left(p^4 + \frac{1}{p^4}\right) + \left(p - \frac{1}{p}\right)$

$= (n^3 + 3n) \times 79 + \sqrt{7}$

$= (7\sqrt{7} + 3\sqrt{7}) \times 79 + \sqrt{7}$

$= 10\sqrt{7} \times 79 + \sqrt{7}$

$= 791\sqrt{7}$

* $x+\frac{1}{x}=n$

$$x^4+\frac{1}{x^4}=\left(x^3+\frac{1}{x^3}\right)\left(x+\frac{1}{x}\right)-\left(x+\frac{1}{x}\right)$$

$$x^4+\cancel{\frac{1}{x}}+\cancel{x}+\frac{1}{x^4}-\cancel{x}-\cancel{\frac{1}{x}}$$

* $x-\frac{1}{x}=n$

$$x^4-\frac{1}{x^4}=\left(x^3-\frac{1}{x^3}\right)\left(x+\frac{1}{x}\right)+\left(x-\frac{1}{x}\right)$$

$$x^4+\cancel{\frac{1}{x}}-\cancel{x}-\frac{1}{x^4}+\cancel{x}-\cancel{\frac{1}{x}}$$

85. If $\left(3\sqrt{x}+\frac{1}{2\sqrt{x}}\right)^2=(3)^2$, $x>0$ then find $x^2(18x^2-7)$?

(a) $-\frac{1}{36}$ (b) $-\frac{2}{63}$

(c) $-\frac{1}{72}$ (d) $-\frac{8}{81}$

$9x+\frac{1}{4x}=9-2\times3\sqrt{x}\times\frac{1}{2\sqrt{x}}$

$\left(9x+\frac{1}{4x}\right)^2=(6)^2$

$81x^2+\frac{1}{16x^2}=36-2\times9x\times\frac{1}{2\cdot4x}$

$\therefore\ 81x^2+\frac{1}{16x^2}=36-\frac{9}{2}$

$\Rightarrow \frac{2}{9}\times81x^2+\frac{2}{9}\times\frac{1}{16x^2}=36\times\frac{2}{9}-\frac{9}{2}\times\frac{2}{9}$

$18x^2+\frac{1}{72x^2}=7$

$\Rightarrow (18x^2-7)=-\frac{1}{72x^2}$

$\Rightarrow x^2(18x^2-7)=-\frac{1}{72}$